PRODUCT ARCHITECTURE / 11 OCTOBER 2026

Human Connection Platform

v0.2 - From discovery to a wanted meeting

Behavioral learning, finite local discovery,
and consent-based real-world coordination

Build the meeting service first.

Explore is the core MVP. Go begins as a bounded event experiment. Prove reciprocal opportunity and real-world outcomes before investing in attraction models or citywide proximity.

Prepared from the founder brief and the supplied 31-page v0.1 architecture. Includes research evidence, product flows, algorithm contracts, privacy design, threat model, data architecture, experiments, and a 12-week plan.

Proposed architecture. Numeric settings are pilot hypotheses unless a source is given. No claim of validated matching performance or guaranteed safety.

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Part 1 - Executive assessment

Go Mode is a plausible coordination advantage, not yet a defensible company. It reduces the mismatch between people who are theoretically dateable and people who can actually meet now. It does not create compatible local supply, establish trust, or predict chemistry. An incumbent could copy the visible feature. A durable business would need unusually effective local recruitment, trusted operations, better reciprocal allocation, and evidence of outcomes people value.

The strongest revised product is a **finite connection service**: a small number of mutually eligible introductions, each tied to a feasible next step. Explore handles future meetings; Go handles short, explicitly chosen windows of immediate availability. Shared event contexts can concentrate supply without pretending that everybody at an event is available for dating.

Five decisions shape v0.2:

- **Keep the boundary, revise the taxonomy.** A preference can be learned; an explicit active exclusion remains binding until its owner changes it. Behavioral contradiction authorizes a prediction update, not a consent update.
- **Replace proximity gradients with public rendezvous.** No exact stranger locations, live distance bands, directional arrows to people, crossing histories, or counts of compatible people. Both users navigate to an agreed venue.
- **Separate selection, experience, reliability, and safety.** Interest is not chemistry; attendance is not integrity; a safety report is not a romantic dislike. Each has different labels, access rules, and consequences.
- **Use overlapping candidate pools, not social castes.** Geography and availability help retrieval. Identity groups and inferred appearance clusters must not become permanent membership boundaries.
- **Prove coordination before personalization.** Start with rules and auditable randomization; use learned models only when they beat a simpler baseline on calibrated, held-out outcomes and an online policy test.

The main founder assumptions I would reject are: finite pools automatically create commitment; private photo embeddings eliminate racial discrimination; coarse distance guarantees privacy; verification makes a person safe; more downstream behavior is always more truthful; and one million registrations necessarily produce enough local training data.

Investment view: This is currently a service and marketplace execution thesis with a possible learning advantage. It is not yet an ML moat. I would fund a bounded pilot with explicit continuation gates, not a broad geospatial platform build.

All numeric caps, timeouts, retention periods, thresholds, sample ranges, and service objectives below are **proposed pilot settings**, unless explicitly attributed to a source. They are not established scientific constants or legal safe harbors.

Part 2 - What problem we are actually solving

The core failure is not simply too many profiles. It is a chain of coordination losses: unclear intention, incompatible availability, unilateral interest, overloaded attention, costly planning, discomfort, and uncertain follow-through. Improving any one stage can help without claiming to solve romantic compatibility.

First-principles question	Assessment	What would change the decision?
What does Go solve beyond ordinary matching?	It can reveal simultaneous willingness to meet and remove days of scheduling. A normal app can add this; it is a feature-level advantage.	More mutually positive meetings per eligible member, after accounting for cannibalized Explore meetings and safety costs.
Does clustering solve overload?	Retrieval clusters improve feasibility. Only finite presentation and capacity rules reduce simultaneous choices. Narrow pools alone merely hide inventory.	Equal-slate-size tests showing local context improves attendance or experience.
Does attraction learning improve relationships?	It can predict selection. That may amplify photo quality and existing bias instead of improving dates.	Incremental improvement in meeting and mutual follow-up outcomes, not just acceptance.
Does mystery help?	It may create curiosity, but withholding identity or suggesting someone is close can create anxiety and pressure. No reviewed evidence validates this exact mechanic.	Voluntary use and comfort in representative usability studies, then better downstream outcomes.
Are we helping or entertaining?	Movement, haptics, and novelty are means. If users check more often without better meetings, the design fails its mission.	Reduced attention cost per positive meeting and sustained outcomes after novelty fades.

Unifying primitive: Connection Opportunity. An opportunity is a time-bounded, consent-governed invitation to a feasible social next step. Its type is PERSON, MEET_NOW, EVENT, or GROUP_ACTIVITY. Its contract specifies eligibility, what information may be disclosed, what accepting means, expiration, capacity consumption, and the next action. A person is not an item to acquire; an opportunity is a proposed interaction.

Do not immediately force all types into one scalar score. An event click, a mutual introduction, and a date are different actions with different denominators. Start with an explicit slate policy: at most a few person opportunities plus one relevant event option. Learn type allocation only after outcomes are comparable and attendance is not confused with romantic consent.

The product should create a finite **attention window**, not a finite social identity. Repeated voluntary group participation can build familiarity. Repeatedly resurfacing someone who declined cannot.

Part 3 - Research findings

Evidence ledger

EVIDENCE-BACKED means the cited research supports the stated observation in its studied population. COMPANY-REPORTED means a product description or company claim, not independent outcome validation. REASONABLE INFERENCE is an engineering or product deduction. UNTESTED HYPOTHESIS requires this product's data. DANGEROUS ASSUMPTION is an unsupported claim that could cause harm if embedded in the system.

Claim	Classification and finding	Limits and design consequence
Attention is asymmetric.	EVIDENCE-BACKED: Pew's 2023 US survey found women more likely to report message overload and men more likely to report insecurity from too few messages; LGB respondents also reported substantial overload. [R01]	Self-report, not an experiment identifying the mechanism. Do not generalize every woman or man, or assume binary heterosexual markets. Measure actual load and reciprocal opportunity.
Large choice sets can worsen decisions.	EVIDENCE-BACKED: Pronk and Denissen found increasingly rejecting choices across successive dating profiles. [R02]	Specific experimental settings and samples; not proof that three introductions is optimal or that scarcity improves relationships.
Online attention concentrates.	EVIDENCE-BACKED: Bruch and Newman describe aspirational pursuit and a hierarchy in messaging patterns in four heterosexual US markets. [R03]	A graph-derived measure of receiving attention is not objective beauty or human worth. Historic platform data cannot justify tiers.
Profiles predict unique chemistry.	Not established: Joel et al. predicted some general liking/being-liked variation, but not pair-specific desire from pre-date traits and preferences in their speed-date samples. [R04]	This does not prove all pair prediction impossible. It invalidates strong claims based only on questionnaires.
Later interaction makes everything predictable.	EVIDENCE-BACKED caution: longitudinal early-relationship research found only modest support for perceiver-by-target compatibility and no ideal-preference-matching effects on interest. [R05]	Post-interaction information may help certain targets; it does not validate a universal compatibility engine.
Familiarity can change selection.	EVIDENCE-BACKED association: couples with longer prior acquaintance showed less matching on observer-rated attractiveness. [R06]	Observational couples sample; selection and recall remain. Repeated community contact is a hypothesis worth testing, not a guaranteed intervention.
Proximity creates connection.	EVIDENCE-BACKED in a different domain: randomized classroom seating affected friendship formation. [R07]	Children and friendship are not adult romance. Transfer only the hypothesis that opportunities for contact matter.

Claim	Classification and finding	Limits and design consequence
Offline communities produce superior relationships.	Unsupported blanket claim. US meeting-pattern research documents online displacement of intermediaries and discusses prior evidence that relationship outcomes need not depend on meeting route. [R08]	Do not market online dating as intrinsically broken or traditional communities as inherently better.
Coarse location is safe.	DANGEROUS ASSUMPTION: security research documents location and personal-data inference in dating apps. [R09]	Hiding exact coordinates in the UI is insufficient. API output, sorting, availability, timing, and repeated queries can leak.
Fewer unanswered chats helps.	COMPANY-REPORTED: Hinge's Your Turn Limits requires responding or ending conversations before additional connections when unanswered load is high. [C01]	Mechanism precedent, not independently established causality for our cap. Do not carry v0.1's percentage-lift claim forward as proof.

The founder's overload chain is plausible but partly unproven. Attention imbalance and frustration have evidence. The complete causal sequence from more options to comparison to lower investment to worse relationships is not established by these sources. Scarcity could also increase pressure, reduce autonomy, and disproportionately harm people with few eligible candidates.

Cross-cultural mechanisms, without romanticizing them

Context	What the research supports	Mechanism to borrow; mechanism to reject
Indian arranged and jointly chosen marriages	Allendorf and Pandian document change and persistence, including greater joint parent-daughter involvement rather than a simple switch to one Western model. [R10]	Borrow explicit long-term intention and optional, member-controlled advice. Reject caste, dowry, family vetoes, and family access to private dating activity.
Nepal, an arranged-marriage setting	Allendorf and Ghimire examine marital quality and spouse choice in a specific social context. [R11]	Autonomy and relationship quality need direct measurement. Do not use low divorce as evidence of happiness; exit can be constrained.
Orthodox Jewish shidduch systems	Qualitative research documents difficulties experienced within matchmaking; community-sponsored research adds context but has its own sampling perspective. [R12, R13]	Borrow a trusted intermediary and deliberate introductions only when voluntarily chosen. Reject communal stigma, status ranking, and pressure to accept.
Chinese parental matchmaking	Research identifies differences between parental and couple objectives and tradeoffs in marital harmony and family outcomes. [R14]	Borrow discussion of practical life context. Do not let parents' economic or lineage goals become the user's optimization target.
Western online and community routes	Meeting routes have changed substantially; shared context is one route among several. [R08]	Preserve independent choice and bridge beyond a person's existing network; avoid treating professional status as compatibility.

These are heterogeneous societies and samples, not national personality profiles. Faith and cultural practices may be voluntarily shared as concrete life preferences. Do not infer religion, caste, ethnicity, or cultural identity from name, face, language, neighborhood, or event attendance. Optional language needs or religious-practice boundaries require explicit user input and a documented policy; no automatic demographic cluster.

The transferable hypothesis is **deliberate, contextual, autonomous introductions with low coordination cost**, not arranged marriage as a superior algorithm. No credible comparative evidence reviewed establishes that fewer alternatives cause more durable, satisfying partnerships across cultures.

Current competitor landscape

Snapshot checked 11 October 2026. Features vary by market, version, and rollout. Official descriptions establish what a company offers or announces, not whether it works.

Product	Verified relevant mechanism	Implication for v0.2
Hinge	Your Turn Limits and a meeting-feedback category in support. [C01, C02]	Capacity management and feedback are not novel. Outperform on allocation and follow-through.
Tinder	March 2026 release describes expanding Chemistry personalization and opt-in Photo Insights. [C03]	Personalization/curation is not a unique pitch. Do not request camera-roll access.
Bumble	Current official materials describe changing conversation/matching experiences; support still lists paid attention features. [C04, C05]	Do not assume the historic women-first mechanic describes every current market. Its roadmap is not a shipped-feature guarantee.
Coffee Meets Bagel	Official communication describes small daily batches; app listing includes Suggested, Discover, and Likes You. [C06, C07]	Finite recommendations can coexist with catalog behavior. Evaluate the entire product, not one screen.
Breeze	Arranges real dates after mutual interest; FAQ describes a limited logistical chat window before a date. [C08, C09]	v0.1's shorthand 'no chat' omits coordination chat. Real-world logistics and venues already have precedent.
Thursday	Current site and app focus on singles events, local discovery, and ticketing. [C10]	Concentrating demand in time/place is a direct competitive alternative to citywide Go.
Happn	Crossed-path discovery; policy describes regular consented geolocation and showing the viewer's crossing point, not the other person's exact position. [C11]	Proximity dating is not new. Do not copy historical crossing traces. No reviewed independent causal evidence establishes why its retention or relationship outcomes succeed or fail.
Meetup	Groups and events built around shared interests. [C12]	Supplies recurring social context; membership or RSVP is not consent to romantic discovery.

Product	Verified relevant mechanism	Implication for v0.2
Eventbrite	Event search, app discovery, and ticket purchase. [C13]	Ticketing can support real inventory. Do not assume a public attendee API or permission to import attendees.
Pokemon GO	Nearby public points of interest, Routes, optional movement tracking, and haptic feedback. [C14-C16]	Borrow public-destination discovery and event cues; reject tracking human targets or rewards for collecting people.
Life360	Trusted-circle sharing with user controls; legal help distinguishes hiding location from ending collection. [C17, C18]	Borrow optional trusted-contact coordination. Ghost must stop this product's collection, not merely hide it from others.

There is no basis to claim novelty for Go's ingredients or an absence of competitors. The proposed combination may feel distinctive; distinctiveness and defensibility are different tests.

Part 4 - Product philosophy

Introduction intelligence, human compatibility discovery. The service should increase the chance of a feasible, wanted encounter and then step back. It should not claim to know a person's destiny, attachment style, morality, fidelity, or social league.

Design commitments:

- Adult-only service; bilateral eligibility and revocable consent at every disclosure and meeting stage.
- No infinite feed, like inbox, public ranking, compatibility percentage, streak, paid exposure, or fabricated scarcity.
- A truthful empty state is a valid outcome. Use 'No available introduction right now' rather than implying the algorithm knows nobody is good enough for the user.
- Identity and enough context precede a meeting decision. Minimal profile information must not mean blind physical commitment.
- Matching feedback stays private. Separate actions control messaging, meeting, disclosure, and reporting.
- Location is a short-lived input for a specific purpose, not a historical user asset.
- No incentives tied to accepting, answering Yes after a date, staying online, or generating more dates.
- Accessibility, non-drinking venues, cost, transport, and permission-denied fallbacks are core logistics.

Gameful interaction means clear, calm feedback when a real action changes state: a single haptic when a valid invitation arrives, a pleasant venue-selection animation, or a public activity route the user chooses. **Manipulative gamification** means manufacturing uncertainty or reward schedules to drive checking. Never animate a radar scan when no fresh inventory is being evaluated; never tell someone to move around to unlock people.

Onboarding and profile

The 2-5 minute goal applies to creating a usable basic profile, not a promise that every age or identity review finishes in that time. Defer optional data; do not hide mandatory checks until after someone has invested effort.

Stage	Minimum collection or action	Reason
Register	Recoverable account credential, adult age eligibility, terms/privacy acknowledgement	Account security and access eligibility; retain birth date in the identity boundary, not analytics.
Discover	Display name, age presentation, photos, who the member wants to meet, relationship structure/intention, broad area, travel willingness, one prompt and a few interests	Bilateral eligibility and an informed introduction. Allow optional boundaries without a questionnaire.

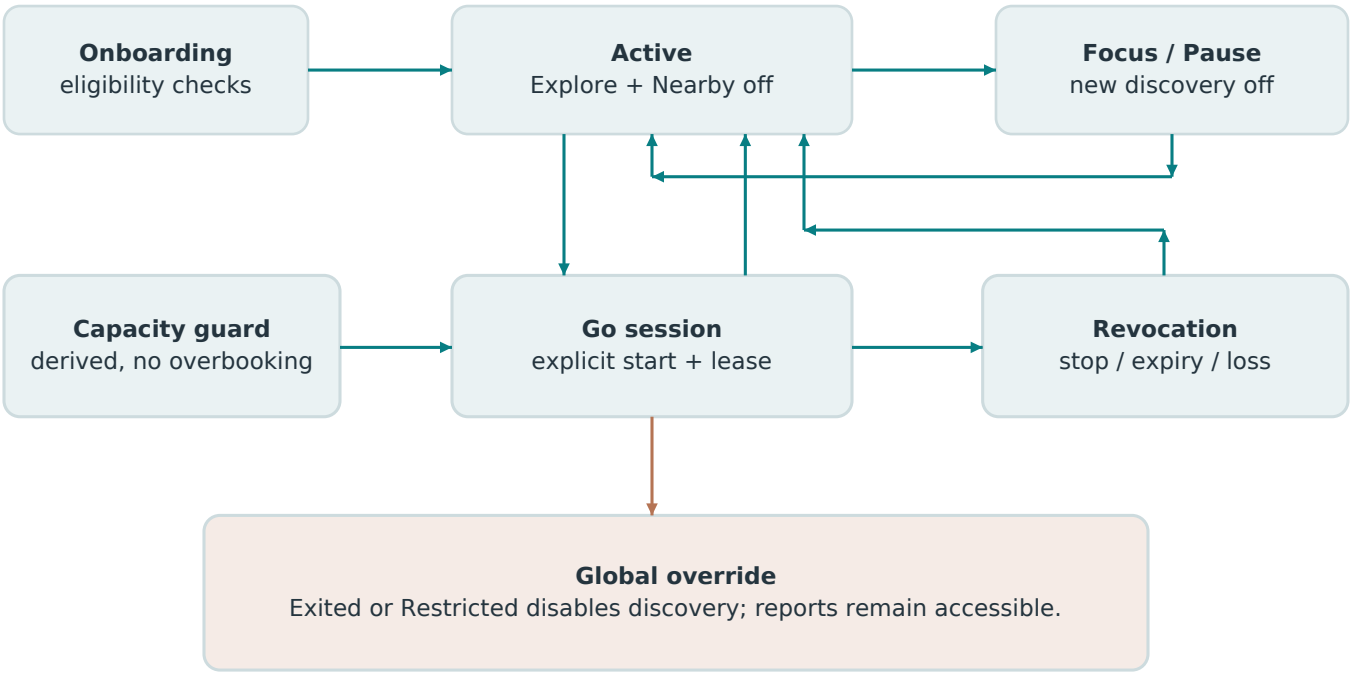
Stage	Minimum collection or action	Reason
Before Go	Free verification appropriate to risk, explicit session consent, public-zone choice, fresh availability, capacity, location notice or approved event check-in alternative	Consent to immediate discovery, not blanket tracking permission. Verification does not certify safety.
Before meeting	Both approve identity card, venue, time, expected activity, accessibility and cost fit; cancel/report access	A clear shared plan and voluntary participation.
Learn later	Availability patterns, optional life goals, privately supplied experience outcomes	Ask when useful to the current action, with skip controls.
Infer cautiously	Contextual accept probability, logistics response probability, uncertainty	Never present an inference as a fact about identity or character.
Never collect for matching	Camera roll, precise history, face-derived race/sexuality/religion, contacts as a social ranking graph, trauma history, 'cheater' labels	Disproportionate risk and weak relevance.

Profile default: two or three current photos, first name or chosen display name, truthful age or a clearly marked age band, relationship intention, a few interests, one short prompt. Date of birth, employer address, home, inferred preferences, report history, and internal probabilities are never shown. Exact adult age is a reasonable default; an age band must never misrepresent age eligibility. Shared-fact explanations are optional and short, not suppressed if they help a user decide.

Part 5 - Modes and state machine

Do not implement one overloaded enum. Use independent dimensions with enforced invariants:

Account: ONBOARDING, ACTIVE, PAUSED, EXITED, RESTRICTED. **Discovery intent:** EXPLORE or FOCUS. **Nearby privacy:** GHOST or GO. **Capacity:** derived OPEN or AT_CAPACITY. **Opportunity:** its own lifecycle. This represents all requested states without implying that Ghost disables ordinary Explore.



User-facing state	Legal entry and exit	Location / discovery / notifications
Onboarding	New account; exit to Active after required eligibility. Can pause or delete.	No nearby processing. No introductions. Verification notices only.
Explore	Active + discovery enabled. Can enter Focus, Pause, Exit, or start Go.	Declared area only; finite future introductions and opted-in event suggestions.
Go	Explicit start from Active/Explore; requires capacity, verification, public context, consent, and fresh presence.	Session-bound nearby processing; actionable invitations only. Returns to Ghost on expiration, stop, permission loss, restriction, Focus, or Pause.
Ghost	Default; set by explicit stop or any Go invalidation. Explore remains independently available.	No GPS requests or nearby lookup, no seeing nearby people, no nearby discovery of this user. Existing messages/plans remain.

User-facing state	Legal entry and exit	Location / discovery / notifications
At capacity	Derived whenever reserved + open capacity reaches the user's limit. Returns when capacity releases.	No new person opportunities or nearby searches. An already reserved opportunity can finish within its lease; plans, chats and generic events continue.
Focus	Explicit choice from Active; return to Explore only by user action.	Go revoked; no new romantic introductions. Existing connections remain; no public exclusivity badge.
Paused	User action from Onboarding/Active; return through eligibility check to Explore + Ghost.	No new discovery or location. Existing messaging may remain by user choice. Plan reminders follow retained plans.
Exited	Explicit closure with optional reason; privacy deletion workflow. Returning requires a new eligibility flow.	No discovery, location, or marketing. Optional separately consented research follow-up only.
Restricted	Safety or eligibility decision; appeal/manual restoration with fresh checks.	Discovery revoked; preserve reporting and appeal. Do not expose reporter identity.

Precedence: Exited/Restricted/Paused overrides all discovery; Focus overrides new romantic discovery; capacity blocks new reservations; Ghost denies every nearby read/write. Server policy is authoritative. Persist `policy_epoch` and `mode_version`; every nearby request, candidate read, invitation, and delivery rechecks them.

Opportunity transitions: DRAFT -> RESERVED -> PROPOSED -> MUTUAL -> PLANNING or MEET_NOW -> RENDEZVOUS_CONFIRMED -> MET_REPORTED -> CLOSED/CONTINUING. PASS, EXPIRE, CANCEL, BLOCK, and RESTRICT can close at earlier stages. Attendance is a reported observation, not an automatic result of arrival GPS. No means no for this pair; changing the radius or mode cannot reintroduce them.

For Go, every consequential transition is compare-and-swap on a versioned record, with idempotency key and expiry. Both venue approvals must reference the same venue/time version. One person's Yes is insufficient. Block or mode revocation wins over concurrent acceptance. A reservation is released exactly once.

Reaching capacity is not consent revocation: acceptance of an existing offer uses its held slot and must not fail because that reservation made the member full. New lookup stops while the allocated opportunity can progress. Explicit Ghost still revokes every unaccepted nearby offer. Reservation-aware checks prevent the last available slot from creating a self-canceling invitation.

Part 6 - Explore Mode

Explore is an asynchronous introduction service for people available over the next several days. Initial hypothesis: a maximum of three active connections and up to three new introductions per week when slots exist. These are caps, not promised deliveries or scientifically optimal values. Offer a lower personal cap; do not punish slower replies or disability-related communication differences.

User flow: short onboarding -> finite introduction card -> private Accept/Pass -> mutual chat and schedule -> choose public venue -> confirm -> meet -> optional private feedback. Photos and sufficient context are shown before acceptance. No swipe stack, rank number, like count, public pass reason, read receipt, or manufactured countdown. An expiry communicates genuine availability limits; request more time when appropriate.

Pipeline: (1) snapshot active status, intent and remaining capacity; (2) apply bilateral explicit constraints and authoritative safety denials; (3) retrieve overlapping local/availability pools; (4) exclude closed pairs and stale profiles; (5) check feasible shared time and both travel budgets; (6) construct time-valid pair features; (7) estimate reciprocal willingness and meeting feasibility, or use rules; (8) allocate capacity with an exploration branch; (9) reserve both slots atomically; (10) recheck eligibility and deliver privately; (11) log exposure and policy; (12) collect stage-specific outcomes.

Availability should support both a concrete shared window and an explicit 'flexible this week' option. V0.1's mandatory shared two-hour window can unnecessarily exclude shift workers and people seeking a short first meeting. A feasibility estimate is useful; a default window length is not a consent constraint unless the member chose it.

On an empty slate, say so without a rejection judgment. Offer the next honest batch time, a relevant public event, or an optional wider travel/time setting. Broaden retrieval within existing permission automatically; broaden a member's hard boundary only when they approve it. Never fill a quota with a pair that fails safety, consent, or minimum feasibility.

Conversation capacity is not inferred solely from who has failed to reply. Reserve one slot for each outstanding bilateral proposal, then count the resulting connection once. A time-limited soft reservation prevents overbooking; delivered invitations also consume exposure budget. A plan inherits its connection slot. Users can close stalled conversations privately; capacity rules must not pressure them to send unwanted messages.

Initial timers to test: 48 hours to respond to Explore, a quiet reminder only if enabled, and an archive suggestion after seven days without progress. Do not automatically record disinterest from expiry. Ended or archived states must be distinguishable from deliberate rejection in training.

Part 7 - Go Mode

Product decision and sequence

MVP positioning: a controlled prototype inside staffed, opt-in events or public venue sessions. Broader neighborhood Go is V1, conditional on demonstrated demand, enough compatible concurrency, and an adversarial review. It is not required for the core MVP to succeed.

Start with Go for existing mutual connections; test new-person Go inside an explicitly dating-enabled event as a separate extension. Existing matches reduce unwanted identity disclosure but do not remove stalking risk. Both modes still require independent opt-in, blocks, current availability, and no person-location output.

Step	User experience	Server condition
Activate	'Open to a short meeting for the next 30 minutes'; public zone/event, current limits, Stop clearly visible	Consent version, server session, free capacity, verification, location permission or approved check-in
Wait	Calm availability card and generic venue/activity map	No user dots, compatible-person count, nearest-first list, or hint to walk toward a person
Offer	One actual introduction with a normal identity card and 'Available to meet in this public area'	Both fresh and eligible; both reserved; for strangers, reveal permitted by session scope
Decide	Private Interested / Pass	No disclosure of unilateral Yes, no reactive proximity updates
Mutual	'You both want to meet'; chat available	Mutual acceptance and current authorization rechecked
Plan	Choose one of a few public venues, expected start, cost and accessibility	Both approve same plan; personal origins never disclosed
Navigate	Each receives directions to the venue	External map receives the venue; app does not send the other person's location
Arrive	Optional 'I'm here' and visual identity confirmation through chat	Arrival is user-reported; no proof of safety or attendance from GPS alone
Finish	End, continue privately, or Something felt off	Stop tracking once discovery ends; venue/time plan remains only as needed

Reject the founder sequence that requires reaching the venue before chat or identity confirmation. A person should be able to ask questions and recognize the intended partner before traveling. Delaying full identity can shift rejection to a more expensive and vulnerable moment. An anonymous invitation may be used for a generic event announcement, but not as a promise that a specific nearby romantic prospect exists.

Revised real-time pipeline

Verify account/session authority and safety eligibility before the geospatial service admits a token. Then check public context and fresh presence; retrieve a bounded eligible pool; reapply pair blocks and bilateral boundaries; check actual current availability and both capacity counters; score reciprocal opportunity; reserve; disclose only permitted card information; wait for mutual interest; agree a public venue; hand off navigation; revoke unnecessary presence; collect private outcomes. Recheck denials at retrieval, reservation, delivery, and acceptance, not only once near the beginning.

Use one fresh offer at a time and an initial exposure budget of at most two new-person offers per session. A proposed five-minute response lease can expire without penalty. Session ends after 30 minutes unless explicitly renewed. Two renewals/hour may be a starting abuse budget, but an open invitation cannot be refreshed by toggling. A short search window with no opportunity should end gracefully; it must not become a recurring mystery notification.

Radius, mobility, and accuracy

Half a mile is approximately 805 meters; it is a search hypothesis, not a guarantee of walkability. A river, highway, closed entrance, mobility need, or unsafe route can make it irrelevant. Use both members' maximum time-to-venue and transport choices. For the pilot, use event/venue scope instead of a circle. For later neighborhood Go, retrieve a coarse catchment and test public destinations within a mutually permitted travel budget, initially around a 10-15 minute journey. Suburban travel can be offered transparently; never silently widen a 'walk now' session to a three-mile drive.

An accuracy radius larger than the relevant public zone makes the location uncertain. Either use an approved event presence check, ask the user to confirm the zone, or decline nearby discovery. Do not tell them a person is nearby based on stale or low-confidence GPS. Indoor floors and apartment-above-cafe situations make coordinates alone insufficient to establish public presence.

Translate Pokemon GO carefully

Official materials verify Routes, points of interest, optional movement recording and vibration, not a public specification of its complete current geospatial backend. [C14-C16] Do not state that a specific S2 level governs today's game without authoritative evidence.

Borrow exploration of public destinations, understandable venue maps, a single event-triggered haptic, and time/place concentration. Reject collectible people, rarity tiers, proximity meters, person-directed routes, movement rewards, and Adventure-Sync-like background histories. The delightful moment is an agreed opportunity to meet, not increasingly accurate pursuit.

Part 8 - Ghost Mode

Use **'Nearby off'** as the primary control label, with Ghost as optional brand language. 'Ghost' can be confused with abandoning a conversation; 'Invisible' can overpromise that previously disclosed information disappears. State exactly: 'You cannot see or be discovered in nearby introductions. Explore and existing chats still work.'

Revocation transaction: increment `policy_epoch`, set nearby OFF, revoke every location-session token, delete the current presence/index keys, invalidate cached pair opportunities, release unaccepted reservations, cancel queued nearby notifications, and publish invalidation to connected clients. Every data-serving path checks the authoritative epoch. An eventually consistent cache alone cannot enforce a hard privacy boundary.

Client immediately stops location subscriptions, geofences used for this feature, retries, and buffered uploads; erases pending location payloads; clears nearby cards. Server rejects late uploads with old epochs. Disable storage in crash reports, request-body logs, background task payloads, and notification metadata. Explore must never quietly recreate real-time location via IP-derived nearby ranking.

Unaccepted Go offers disappear with a generic unavailable message. A mutual connection can remain a normal conversation. A confirmed rendezvous remains a plan unless the member cancels it; show a separate 'Stop nearby and cancel this plan' action. Neither person is told that the other entered Ghost or where they went. No automatic restart when they leave a venue, reopen the app, or regain permission.

Distributed-systems limit: if the device is offline when Stop is tapped, it cannot synchronously revoke server knowledge. Stop locally at once, send revocation when reachable, and display a clear pending/offline state until acknowledgement. Bound the server's residual authority with a short presence lease (pilot 90 seconds), not the 30-minute session maximum. A policy acknowledgement means no future authorized disclosure; it cannot recall a card, screenshot, or push already delivered. Push bodies should therefore never contain identities or location clues.

Measure online revocation acknowledgement latency, stale-card retrieval attempts, queue invalidation, and presence TTL expiry. Pilot service objective: online revocation acknowledged within two seconds at p99; no subsequent accepted reads under the old epoch. During authorization-store failure, deny nearby discovery. Test stop versus accept races, clock skew, retries, offline mode, worker restarts, and backup restoration.

Part 9 - Geospatial/privacy architecture

A system without a historical location database

Yes: candidate proximity can operate on only current, expiring presence. This does **not** make the service anonymous to the operator, nor erase location-related facts such as a mutually agreed venue and date. Distinguish ephemeral raw location from disclosed rendezvous records and explicitly submitted incident evidence.

Default event architecture: member selects a supported public session, completes an event check-in, and sends only a short-lived event token. A host-issued rotating check-in code increases abuse cost but can be relayed and is not cryptographic proof of physical presence. For a staffed pilot, host assistance is the practical fallback. No GPS is necessary for this mode.

Later neighborhood architecture: device obtains a current fix only during Go -> local accuracy and public-zone checks -> encrypted upload to a dedicated presence process -> server validates session/zone and computes a coarse cell or zone membership -> discards the raw fix -> ephemeral geospatial index -> opaque pair eligibility -> main coordinator. Keep only the latest coarse record with a 90-second lease and no disk persistence, append-only file, snapshot, swap, or request-body telemetry. Do not put raw coordinates in the job queue or analytics lake.

Where exact transient computation is needed to confirm a zone, treat raw coordinates as sensitive plaintext in process memory, not 'encrypted computation'. Target discard immediately after the request and enforce a short processing deadline, proposed ten seconds. A compromised live process can still read them. Separate process credentials and network policy reduce the blast radius; encryption at rest does not defend against an authorized but compromised application.

Spatial implementation: PostGIS for static public polygons and venues; an ephemeral Redis-compatible service for session tokens and zone membership only when neighborhood Go warrants it. H3 is an optional internal candidate index. Resolution 8 has average hexagon area about 0.737 km2, resolution 9 about 0.105 km2; size varies, and an index is not an anonymity guarantee. [T01] Prefer broad multi-block public zones over fine cells. Retrieve adjacent cells and check polygon/distance uncertainty to avoid boundary misses. A cell ring is not walking distance. Never expose cell IDs, distance sort order, or target-specific presence endpoints to clients.

Threat-informed output design

Proposed control	What it helps	What it cannot promise
Coarse cells	Reduces stored precision and query cost	Repeated disclosures or a sparse cell can still identify a location
Rotating session IDs	Limits accidental linkage in logs and index access	Identity cards, timing, and colluding accounts can relink people
Fixed response epochs and sticky session zone	Reduces movement-based probing and timing changes	Not formal privacy; known identities can still imply presence
Population threshold	Suppresses especially sparse public disclosure	Bots and collusion can manufacture crowd size; k users are not k anonymous possibilities
Random threshold/delay	May reduce a particular side channel	Repeated noise can be averaged; stale signals hurt coordination; not the primary defense
Public-zone allowlist	Avoids arbitrary activation at homes	Mixed-use buildings and sensitive venues require review; does not prove a person is safe
Rate/query budgets	Raises enumeration cost	Device/IP checks alone cannot stop a determined attacker
No person distance/count/history	Removes strong observation channels	A voluntarily revealed local card still reveals coarse current participation

The pilot avoids pre-mutual compatible-person counts entirely. Venue selection uses a small zone-level roster, not a continuously optimized exact midpoint; otherwise venue changes themselves can become a location oracle. Requests cannot specify an arbitrary target ID, arbitrary search origin, or unlimited filters.

Before broad Go, use a conservative public-crowd threshold (example 20 recently verified session participants) plus individual consent and query controls, not as a claim of k-anonymity. If thresholding excludes small LGBTQ or minority pools, offer staffed events or existing-match rendezvous; do not weaken safety or publish small segment counts. Review whether anonymous discovery is needed at all.

Mobile behavior and operating cost

Android limits background location and imposes service/permission requirements; Apple may suspend apps and supports explicit background-location mechanisms. Accuracy and update frequency affect battery use. [T02-T04] An app cannot promise continuous fresh background presence across all device states.

MVP is foreground/event-bound. Later a user-started, visibly active OS-supported session may continue briefly in the background after native testing and store-policy review. Never silently request Always permission. Prefer coarse permission, motion-triggered updates, and a heartbeat that refreshes availability without always refreshing GPS. Use a new fix only when movement or staleness requires it. Navigation belongs to the user's chosen map app after plan confirmation; nearby discovery can stop.

Prototype budgets: 30-60 second presence heartbeat, latest-fix maximum age 90 seconds, no more than one location acquisition per minute while stationary unless needed to resolve uncertainty. These budgets are experiments, not guaranteed platform behavior. Benchmark battery on real older devices, approximate-only

permission, poor reception, and background throttling. If freshness lapses, expire eligibility rather than extrapolate movement.

Load depends on concurrent sessions, not registrations: 1,000 concurrent sessions with 30-second heartbeats generate about 33 writes/second; 100,000 generate about 3,333/second. Score on meaningful state changes or bounded batches, not every fix. Cache public venue route estimates by coarse origin and transport, not identifiable raw journeys. Pilot p95 offer generation can tolerate several seconds; a 30-second heartbeat does not provide subsecond proximity. Budget for maps, verification, support, and venue operations as well as compute.

Legal design boundary

For a proposed US pilot, review launch-state privacy, consumer-protection, age-assurance, biometric, and event obligations before selecting vendors. Illinois BIPA has notice/release and retention/destruction duties for covered biometric processing; a vendor does not remove the operator's obligations. [L01] CCPA identifies precise geolocation, sexual orientation, and certain biometric/inferred information as sensitive, subject to applicability and detailed rules. [L02] FTC location-data actions are concrete warnings against hidden secondary use; they are not a universal permission checklist for dating apps. [L03]

For EU expansion, map lawful bases, special-category processing, privacy by design, data-subject rights, and DPIA requirements under GDPR; orientation and some inferences deserve particular care. [L04] Complete jurisdiction-specific assessment, vendor contracts, data-flow review, and deletion validation. No claim here that consent alone, a short TTL, or use of a vendor makes the service compliant.

Part 10 - Cluster architecture

Use **overlapping, expiring candidate pools**. A user does not belong to one hidden social class. Candidate membership is a query result over permitted geography, bilateral eligibility, time, transport, and optionally an explicitly joined event. Do not cluster by race, caste, inferred ethnicity, attractiveness, wealth, employer prestige, or a latent score that reproduces those boundaries.

Separate three objects: (1) a spatial index for fast retrieval, (2) a context pool such as a public event, and (3) a finite presentation budget. A pool may contain hundreds of eligible users while the user sees only a few introductions. That is a policy, not deception, provided the product explains curation and does not invent scarcity.

Retrieval and expansion algorithm

1. Build `eligible(u, v, t)` from active status, blocks, explicit bilateral constraints, discovery scope, and current capacity. Access-sensitive exclusions belong here, not in learned ranking.
2. Retrieve local public-zone or declared-area neighbors and compatible time windows. For Go, require current session leases; for Explore, use non-live areas and future availability.
3. Form overlapping pools: local logistics pool, opted-in event pool, and an adjacent-area exploration pool within the member's existing travel permission. Do not require all soft interests to match.
4. Measure eligible edge count, predicted opportunity mass with uncertainty, recent coverage, and unseen candidates. Counts are internal; never expose small sensitive populations.
5. If local supply cannot sustain the next finite slate, expand geography/time within authorized boundaries. Keep a small bridge allocation even when local supply is adequate, to test retrieval blind spots.
6. Ask before changing hard distance, intention, age, or other active boundaries. Offer an event or an honest empty slate when no acceptable expansion exists.
7. Deduplicate across pools, exclude passed/blocked/closed pairs, and recompute before reservation. Do not show the same person through an event merely to circumvent a previous No.

Pilot default: retrieve up to 100-200 eligible neighbors per member where supply exists; this is a computational bound, not a quality threshold. At small scale, score all feasible edges. Store candidate-source tags only long enough to audit retrieval; membership itself expires with its context.

Method	Decision	Why
PostGIS spatial indexes and time joins	MVP	Clear queries and debuggable eligibility, enough for one market
H3 or S2	Optional V1 indexing; choose one	Fast partitioning/neighbor lookup; neither solves privacy or social compatibility
Dynamic geofences	Use curated public polygons and event windows	Catchments adapt to actual walking access; avoid following a moving user with an exposed radius

Method	Decision	Why
DBSCAN/HDBSCAN	Offline operations analysis only, if needed	Useful for understanding aggregate activity density; unstable clusters and sensitive location history are poor foundations for identity assignment
Graph clustering	Defer	Reveals communities but can reproduce homophily, segregation, and past exposure; audit before any retrieval use
Learned two-tower retrieval	Only when exhaustive eligible scoring is a measured bottleneck	Embeddings must improve retrieval recall without excluding difficult-to-serve users

Evaluate candidate recall before ranker quality. Sample allowable edges outside the primary pool offline, estimate how many viable opportunities retrieval misses, and examine who bears the loss. Use consenting audit demographics only in a separated evaluation environment. Never expand into an exclusion just to satisfy fairness.

Density is local and time-dependent. Illustrative decomposition: 1,000 local registrations x 40% weekly active x 10% simultaneously in Go x 25% bilateral eligibility x 50% available capacity yields only five potential candidates before reciprocal interest or travel barriers. These factors are hypothetical, dependent, and not a forecast. Under a crude Poisson model, probability of any eligible candidate is `1 - exp(-lambda)`, but real clustered attendance violates homogeneity. Measure the empirical distribution by time/place, including the fraction with zero feasible edges. Events raise concurrency more directly than a larger citywide registration count.

Part 11 - Preference taxonomy

There is no universal ranking of words versus actions. Authority and predictive information are different dimensions. A consent boundary is authoritative even if a model thinks the member might enjoy violating it. A durable outcome is meaningful for relationship formation but too delayed and selected to replace every early-stage label.

Class	Examples	Authority and learning
Structural eligibility	Adult age, bilateral who-to-meet compatibility, account restrictions	Non-negotiable gates. Use self-declared identity/eligibility, never infer orientation from faces or behavior.
Safety/consent boundary	Block, rejection, discovery consent, current unwillingness	Absolute for the active scope. Learning cannot override.
User-controlled exclusion	'Do not introduce smokers'; maximum travel; explicit relationship-structure requirement	Binding while enabled, regardless of whether product staff consider it a 'real' dealbreaker.
Strong but flexible preference	'I usually prefer someone who does not smoke; other introductions are OK'	Strong prior within permitted candidates. Editable; behavioral evidence may update predictions.
Soft preference	Interests, ordinary activity preferences, flexible age preference within adult hard range	Weak/moderate prior; infer conditional effects with uncertainty and exploration.
Self-description	Lifestyle, goals, availability, personal interests	Claimed attributes with freshness/provenance, not independently established truth.
Selection behavior	Accept, pass, respond	Evidence about a particular decision under a displayed context.
Experienced outcome	Felt attraction, wanted another meeting	Person-specific report about a particular encounter. Keep both sides separate.
Durable outcome	Ongoing connection, success exit	Optional, censored evidence about longer-term outcomes; never assume silence means failure.
Reliability/integrity/safety	Attendance, profile discrepancy, harassment	Separate operational systems; no fused moral or desirability score.

An ordinary preference has `stated_value`, `stated_strength`, `flexibility`, `source`, `updated_at`, and `consent_scope`. Learned state has a posterior distribution, evidence count by label type, effective sample size, recency, context coverage, and model version. Never overwrite the user's original statement with a learned value.

Treat relationship intention explicitly: 'open to exploring' is not the same as 'only monogamous long-term partnership'. If either member requires a structure or goal as a hard exclusion, it gates. If both expressly allow uncertainty, model it as context; do not decide for them that attraction should override it.

Part 12 - Dealbreaker redesign

The founder correction is useful only if implemented as **clear choice at the time a boundary is created**. 'Dealbreaker' cannot mean hard in the interface and soft in the backend.

For each supported preference, offer: **Only introduce within this boundary**; **Strong preference, but open to others**; **No preference**. Default ambiguous legacy dealbreakers to the hard setting during migration and let the member review. Do not silently migrate v0.1 exclusions into flexible priors.

Case	Correct behavior
A user explicitly excludes smokers	Never recommend a known smoker. Accepting someone before learning they smoke does not authorize an override.
A user allows exceptions and accepts several smokers	Update the conditional selection model cautiously; do not infer that smoking caused the acceptance.
An existing connection reveals a conflicting trait	Let the user choose whether to continue this specific connection; one exception is not blanket permission to alter future filters.
A preference seems restrictive and supply is empty	Offer neutral settings review: 'Want to review your preferences or travel range?' No pressure and no claim that nobody wants them.
A safety-related fear is expressed through a proxy	Respect the explicit boundary if supported; explain the observable conduct controls available. Never infer trauma or predict infidelity.

A safe preference-review prompt should not disclose or overinterpret another person's traits. Prefer 'Would you like to broaden this preference?' over 'Your dates prove you like smokers.' Only suggest a behavior-based explanation if the evidence is actually supplied, consented, and sufficiently supported; the member can disable such prompts.

Learning contract: use contrary behavior only from permitted observations. If the hard policy never exposes out-of-boundary candidates, the system has no evidence about them. Do not fabricate counterfactual preference shifts from a narrow policy's data. Do not use hidden out-of-boundary exploration to collect that evidence.

Persist changes with timestamp, source USER, prior version, and explicit scope. A single-pair exception, if supported later, expires with that connection and is never generalized. Blocks and safety exclusions do not support algorithmic exception prompts.

Part 13 - Behavioral-learning system

Three learning tracks plus operational reliability

Selection predicts what a member will choose from an actual introduction. **Experience** predicts their reported outcome after meeting. **Integrity/safety** routes conduct concerns to a separate process. Reliability predicts logistical follow-through only from contextually appropriate observations. They may share identifiers through controlled joins; they must not share a universal score or unrestricted training table.

Signal	Meaning and noise	Update, confidence, decay, and bias
View / details opened	Delivery/exposure or curiosity, not attraction	Use to determine observation opportunity. No positive romantic label; placement/accessibility bias.
Dwell	Reading speed, distraction, screen state, or interest	Exclude from MVP targets; later diagnostic only. Avoid duration optimization.
Explicit pass	Declined this card at this time	Binary selection label only after viewed; context-specific and not a judgment of the other person. Recent observations may be more relevant.
Explicit accept	Willingness to connect, not physical attraction	Selection positive; shrink user-level effects. Model indiscriminate acceptance through actor intercept, never reward it.
Timeout / ignored card	Unknown preference, competing life demands	Censored/undecided state; not a negative preference label. Separately evaluate delivery and availability.
Mutual	Joint willingness under this disclosure policy	Pair-level outcome, not two independent new labels in addition to the same decisions.
Message / reply	Interaction initiated or returned	Event/latency target with observation window; no content mining or reply-speed moral score.
Conversation length	Could indicate interest, difficulty, or endless chat	Do not use as success or attraction target.
Meeting plan	A shared commitment with possible cancellation	Scheduling target; distinguish expired proposals and completed plans.
Attendance	Self-report, bilateral confirmation, or disputed evidence	Separate outcomes for both/one/unknown; no GPS-only proof. Context-sensitive reliability update.
Post-date Yes / No / Unsure	Experienced desire to meet again	Categorical label with optional abstention, not 1/0/0.5. Keep each person's answer private.
Second meeting	Reported follow-up, possibly off-platform	Distinct delayed outcome with censoring. No booking does not refute a Yes.

Signal	Meaning and noise	Update, confidence, decay, and bias
Success exit	Self-reported meaningful ongoing connection	Competing-risk outcome with missingness; optional 30/90-day follow-up, no forced couple confirmation.
No-show / contradiction / safety concern	Possible logistical failure, integrity issue, or harm	Route separately with provenance and dispute/appeal; never a generic romantic negative.

Tractable probabilistic design

For directional selection, begin with a pooled regularized logistic model:

```
logit p_select(u -> v | x, mode) = alpha_mode + b_u + beta_mode . x_uv + delta_u . x_uv
```

`b_u ~ Normal(0, sigma_b^2)` represents selectivity; `delta_u ~ Normal(0, Tau)` permits heavily shrunk individual deviations only after evidence. Pair features are pre-decision values: permitted stated fit, travel/time feasibility, a small set of optional common interests, profile presentation version, and current context. Do not add a permanent candidate popularity intercept in MVP. Audit whether richer pair features later reproduce a global hierarchy.

The prior for `delta_u` may reflect a stated flexible preference, but fit its scale on population data, not an arbitrary 'importance 3 equals twenty swipes' rule. Use hierarchical partial pooling; learn each stage independently at first. A long-lived per-user model is not warranted by three decisions or five dates. Versioned batch fitting is easier to inspect than v0.1's approximate online Newton update.

For experience, start with a population categorical model of Yes/Unsure/No conditional on an attended date, with missing feedback handled separately. Do not stack future feedback from the target encounter into its own pre-introduction feature vector. Later history features must contain only earlier interactions and carry timestamps. For durable outcomes, survival/competing-risk analysis is more appropriate than declaring 'no relationship by day 30' a failure.

Drift: compare time-windowed validation and candidate half-lives such as 30/90/180 days; select decay by predeclared out-of-time performance and stability, not narrative intuition. Until then, preserve timestamps and use a modest bounded recent window. Changes in intention or availability trigger feature refresh and invalidate incompatible opportunities; reset only relevant learned state, not blocks or safety records. Consent never decays automatically.

Bias and logging

Observed behavior is conditional on eligibility, retrieval, display, decision, attendance, and feedback participation. Propensity logging addresses only the parts actually randomized with known support. It cannot recover chemistry for excluded pairs or eliminate unmeasured reasons people meet and respond. This distinction follows the selection-bias problem formalized in recommender research. [M01]

Log candidate policy version, feature snapshot/version, retrieval source, decision history, placement, model version, randomization branch, exact conditional choice probability, remaining capacity, and experiment assignment. Do not log raw location, report text, or sensitive audit demographics in the learning stream.

For a sequential randomized allocator with choice set F at step j, use `q_j(e | history) = (1-epsilon) * I(e is rule-best) + epsilon / |F|` if the exploration branch is uniform over that exact feasible set. Record F's

reconstructable identifier, size, seed, step, and chosen probability. Later steps depend on earlier choices; this is a conditional decision probability, not a pair's marginal chance of being shown in the whole slate. Do not label it as the latter.

Candidate sampling adds another probability term if randomized; deterministic excluded edges have zero support. Complete slate probabilities are products of conditional probabilities and often unusably small. Full marketplace counterfactual evaluation also faces capacity/interference effects. Use IPS/SNIPS or doubly robust estimators only for supported estimands, with clipped-weight sensitivity, effective sample size, and uncertainty; confirm policy changes online. A human changing a batch creates a separate logged policy with unknown propensity, excluded from causal replay unless the intervention is modeled.

Part 14 - Attraction-learning architecture

Define the target before naming the model. `P(accept card)` is selection. `P(reports felt attracted | met)` is experienced attraction. `P(mutual desire to meet again | met)` is broader experience. Neither is an objective attractiveness score. Do not describe the first as learned chemistry.

MVP learns none of these per individual. It records decisions and optional experience labels. V1 may learn pooled, interpretable selection. V2 may test low-dimensional collaborative representations after sufficient exposure diversity, privacy assessment, and comparison with nonvisual baselines. Image features are optional research, not an inevitable stage.

Component	Target / labels / features	Entry gate and baseline	Bias, calibration, cold-start and uncertainty
Pooled selection	Explicit accept among viewed, decided cards; stated/context pair features	Pilot estimate: 5,000-20,000 decisions with adequate outcomes across contexts; beat rule ordering and segment base rate	Display/presentation and exposure bias; Brier, log loss, calibration slope/intercept and slices. New users get pooled prior; uncertain scores approach baseline.
Individual selection residual	Same target; user's earlier permitted decisions and small shared-feature interactions	Often 50-100 diverse decisions before even testing a residual; a research starting range, not sufficient proof	Effective support and learning curves govern. Avoid type inference from three similar choices. Reset/disable available.
Experienced attraction	Optional post-meeting attraction tag or direct opt-in research label; earlier context only	Several thousand attended meetings with enough attraction-label responses; beat overall/context rate	Optional positive chips are selectively observed, not complete negative labels. Ask a balanced research question only with consent; use categorical/missing states.
Collaborative pair embedding	Accept/meet targets; anonymized-in-access interaction graph and permitted content	Tens/hundreds of thousands of diverse decisions, stable time-split gain over regularized models	Cold-start content fallback, memorization and popularity audit, uncertainty-aware exploration. No user-neighbor identities exposed.
Optional visual representation	Incremental selection prediction from opted-in profile imagery	Research only after above; no universal label threshold; demand meaningful held-out and online downstream gain	Face/skin-tone, age, class, disability and photo-quality proxies; never infer race labels. Consent, deletion, counterfactual tests, audit and rollback required.

The sample ranges are budgeting hypotheses. Deployment requires learning curves, calibration precision, held-out users, and enough minority/negative outcomes; there is no scientifically justified universal '10,000 dates unlocks image AI' threshold. A label count without variation can be nearly useless.

Visual-model boundary: do not build face recognition for attraction, infer race or identity categories, or use embeddings from the verification vendor. Verification images and vectors never enter recommendation training. General visual embeddings can still encode protected traits even without named labels. Removing an ethnicity column does not remove discrimination. A probe finding no race signal is not proof of fairness; it may simply be weak.

If a later research program earns approval, use the smallest frozen representation necessary, restrict it to consented profile photos, remove metadata/background location cues, compare with photo-quality-only and nonvisual models, audit sensitive-proxy predictability in an isolated consented dataset, and measure opportunity distribution. Test new photos of the same person to detect presentation confounding. Require an improvement in wanted meetings or mutual follow-up, not just a higher accept rate. If benefit is small or audits fail, do not ship it.

Exploration: a proposed 10% of eligible opportunity allocations can use constrained randomization during the pilot; choose the percentage through cost/coverage review. Never deliberately cross a boundary or expose someone to a safety risk to learn. Do not combine Thompson sampling and fixed exploration until you can explain total exploration and compute/log the actual serving policy. Users can review/reset learned personalization without losing structural exclusions.

Part 15 - Reciprocal matching

Targets and correct probability factorization

Let X include only information available at invitation time. Let A and B be the two accept decisions; M an attended meeting; Y a mutually positive continuation outcome. Then a valid chain is:

```
P(Y and M and A and B | X) = P(A | X) * P(B | A, X) * P(M | A, B, X) * P(Y | M, A, B, X)
```

This is a probability identity with correctly conditioned events, not a claim that each term can be reliably learned. Blind simultaneous decisions prevent influence from seeing a Yes; they do not prove conditional independence because unobserved shared context remains. Do not multiply separately trained unconditional probabilities or double-count scheduling and attendance.

Prefer a pooled **joint acceptance model** when label volume supports it, trained on pair proposals with both decision states and missingness distinguished. A four-outcome model for Yes/Yes, Yes/No, No/Yes, No/No can capture dependence; undecided cases need a separate response process or censoring treatment. Until then, a product of directional calibrated estimates is a transparent approximation with calibration checked at the pair level. The founder's 0.90 x 0.03 versus 0.55 x 0.55 example illustrates reciprocal imbalance, but its numeric joint probabilities require that assumption.

At launch, stated-fit arithmetic is a **priority heuristic**, not a probability. Human-reviewed logistics plus eligibility and randomized ties are enough. Do not display or silently interpret a scaled 0-1 rule score as 80% likelihood.

Use a lexicographic policy: eligibility and safety gates first, enforce capacity and coverage guardrails, then maximize estimated wanted meetings. In the early pilot, meeting completion is the observable optimization target while feedback is an outcome check. Once enough labels exist, target mutually wanted continuation after meeting, with attendance as a constrained diagnostic. Do not permanently reward any first date with arbitrary utility weights `v1=1, v2=3`.

Model contracts

All learned components must pass temporal and user-disjoint checks where relevant, compare to a base rate and simple regularized baseline, and report Brier/log loss plus calibration by mode and meaningful segments. Use bootstrap/clustered confidence intervals; AUC alone is insufficient.

Model	Target and labels	Features and starting data requirement	Failure/fallback
Response	Responds within a disclosed window among delivered offers; expiry distinct from pass	Delivery channel, local time, session freshness, earlier aggregate response history; a few thousand matured offers	Accessibility/work-pattern bias; never equate slow with bad. Use mode base rate when uncertain.
Joint mutual interest	Both explicitly accept same proposal	Directional features, disclosure policy and context; roughly 2,000-5,000 mature pair proposals as initial fitting budget	Dependence and censoring; fallback to rules or a validated approximate product.

Model	Target and labels	Features and starting data requirement	Failure/fallback
Explore plan/meeting	Attended meeting by defined horizon, conditional on mutual	Shared availability, travel bands, lead time, venue accessibility/cost fit; hundreds for base rates, thousands for multivariable model	Off-app meetings/missing attendance; separate unknown. No punitive individual model.
Go meet-now	Both meet during agreed window, conditional on mutual immediate consent	Freshness, event/zone, time-to-venue bands, plan acceptance; dedicated Go labels, not copied Explore coefficients	No location-history features. Sparse Go data uses context base rate; no generalization claim.
Attendance	Both report attendance, one reports, disputed, unknown; conditional on plan	Lead time, cancellation timing, transport/context, established operational history; initially rule/base rate	False reports and disability confounding. Human review before sanctions; no blanket 'two strikes'.
Experience	Each Yes/Unsure/No after a reported meeting; joint continuation distinct	Pre-meeting permitted features and prior experience summaries; thousands of attended pairs before useful multivariable testing	Missing-not-at-random and reporting style; pooled categorical prior, abstain from personalization.
Durable connection	Optional ongoing connection at 30/90 days, with exit reason and follow-up censoring	Prior consented outcomes, elapsed observation time; multi-month cohorts, usually population-level even at large scale	Nonresponse and survivorship; competing-risk analysis, no personalized promise.

Shared services: eligibility, explicit constraints, profile validation, capacity reservations, basic pair features, feedback schemas, and audit tooling. Shared models only when tests show transfer: selection may use a common backbone with mode interactions and separate calibration. Keep Go availability, Explore scheduling, attendance contexts, and notification policies distinct. Never let a Go acceptance overwrite a long-term intention setting.

Uncertainty: missing history is not low quality. Use pooled priors, uncertainty intervals, new-user exposure allocation, and conservative claims. Avoid a blanket lower-confidence-bound ranker that systematically suppresses new users. Exploration handles uncertainty within feasible, consented candidates; safety uncertainty routes to review, not an exploration bonus.

Part 16 - Capacity/exposure fairness

Use a **general graph maximum-weight b-matching formulation**, not a forced men-versus-women bipartite graph. Users may be eligible for overlapping sets across gender and orientation. A min-cost-flow implementation is appropriate only for a genuinely bipartite subproblem; it is not a universal replacement for general-graph b-matching.

For feasible edge e=(u,v), choose `z_e in {0,1}`. Maximize `sum(z_e * value_e)` subject to `sum(e incident to u) z_e <= free_slots_u`, per-session/per-period exposure ceilings, one live proposal per pair, and explicit safety/consent gates. Additional waiting-time priorities and soft exposure deficits promote coverage. A hard minimum introduction for everyone can be infeasible; report unmet demand and use slack, never violate consent to satisfy a floor.

MVP uses an auditable greedy allocator with waiting-time rotation and randomized ties/exploration. Benchmark it against an exact small-graph solver offline to quantify lost allocation value. Move to a general-graph solver when measured benefit exceeds complexity. Stable matching addresses stability under assumed preferences, not automatically our objective, consent uncertainty, or capacity fairness. It is not the default merely because dating sounds like marriage matching.

Attention accounting: bound both active connections and proposed exposures. A high-demand person cannot receive hundreds of likes because there is no unilateral inbox and a proposal reserves both sides. Each profile has an exposure budget independent of incoming demand. Close/reject releases capacity but does not erase the exposure ledger or permit a new account to evade a block. Do not require replies to harassment to unlock capacity.

Group or situation	Expected benefit	Risk and correction
High inbound load	Fewer simultaneous decisions and dead conversations	Too tight a cap can feel controlling; user can lower capacity, close connections, or pause without penalty
Low reciprocal opportunity	Fair exposure and better logistics may help	Greedy reciprocal scoring can hide them permanently; use wait-time coverage, new-user priors, event alternatives and retrieval audits
New users	Pooled priors and reserved exploration support entry	No negative treatment for absent history, no 'reliable dater' badge that stigmatizes newcomers
Small LGBTQ/other eligibility pools	Overlapping retrieval avoids unnecessary fragmentation	Public safety thresholds can reduce Go availability; staffed events and Explore remain accessible
Users with constrained mobility/time	Travel and availability fit prevents infeasible offers	Do not penalize response speed or require walking; offer flexible accessible meetings

Audit the full funnel: eligible supply -> retrieved edges -> allocated exposures -> mutuals -> meetings -> experience. Compare coverage, concentration, waiting time, false-negative estimates, and calibration conditional on opportunity, while also reporting absolute exclusion. Eligibility itself can reflect geographic segregation or discriminatory product filters; it is not a morally neutral denominator.

Sensitive audit attributes are optional self-reports with separate consent, access and minimum reporting sizes. Never infer race from photos to fill the audit dataset. Sparse intersectional slices need uncertainty and qualitative research, not a claim of parity. Equal exposure cannot force equal attraction; the platform is responsible for controllable opportunity and disclosure policies, not engineering romantic acceptance quotas. Give members a way to challenge settings or reset personalization without exposing others' private responses.

Part 17 - Feedback engine

Feedback is a short **interaction-learning step**, not another onboarding questionnaire. It is always optional and never a condition for new introductions, refunds, reporting, or account deletion.

First ask, when needed, **'Did you meet?'** with Yes / No / Prefer not to say. After a reported meeting, typically the next day at a user-appropriate time: **'Would you like to see this person again?'** Yes / Not sure / No / Skip. One optional follow-up can offer at most two reason chips. No repeated nagging and no public rating.

Route	Suitable prompts	Storage and consequences
Matching	Easy conversation, felt attraction, shared outlook, no spark, different goals, different lifestyle	Private encounter-level preference evidence; no public labels. Optional chips are missing when not chosen, not automatic negatives.
Integrity	Something important differed from the profile	Separate optional discrepancy form, evidence and review. A single mismatch is not proof of intentional lying.
Safety	I felt uncomfortable / Something felt off	Immediate independent support path. Ask whether they want to block, report, or get help; do not require categorizing it as a crime.

If discomfort might reflect awkwardness rather than misconduct, let the member choose whether to provide a matching response, a safety concern, both, or neither. Do not automatically train attraction on the safety disclosure. Reporting should require as little detail as possible; operational triage can request more later.

Something felt off is available from profile, chat, plan, post-meeting card, settings, and after unmatching. It opens: immediate safety/help option; optional description; block independently; report independently; optional plan sharing/help resources; acknowledgement and case tracking. Preserve access after the other account disappears. A report can be filed without blocking, and a block without reporting.

Feedback itself is not communication consent. A private Yes must not automatically be revealed, even if both answered Yes. Provide a distinct 'Invite them to meet again' action. The other person never receives a negative answer, reason chip, model update, or explanation of why a connection ended. 'Private' means protected access, not a promise that lawful data requests can never reach it.

Attendance provenance: BOTH_REPORTED, ONE_REPORTED, HOST_CONFIRMED_FOR_EVENT, DISPUTED, UNKNOWN. Event admission is not proof that two people interacted. Optional paired check-in codes can support meeting evidence but are gameable; they must not become mandatory location proof. The system should not infer a no-show from silence or punish someone who left for safety.

Do not down-weight a Yes because no second date was booked in 21 days. The date may have happened off-platform or logistics may have intervened. Keep declared experience, later behavior, and missingness as separate observations. For durable outcomes, offer an optional success-exit reason and separately consented follow-up; quitting is otherwise unknown.

Part 18 - Safe Zones / real-world meeting

Call these **approved public meeting places**, not guaranteed Safe Zones. A busy cafe can still host harassment; a partner badge must not imply supervision or an emergency-response promise. Publish the specific checks and when they were last reviewed.

For each venue maintain: public address and entrance, opening hours/timezone, live session eligibility, accessible entrance/seating/toilets where verified, seating/no-purchase options, cost band, alcohol-free options, staff contact and training status if partnered, transit access, complaint history, emergency exit notes, and review expiry. Require human approval at launch. Do not treat map ratings, police statistics, or a school's reputation as a safety certification.

Exclude private homes, hotel rooms, isolated trails, closed venues, sensitive service locations, and places whose access conflicts with the member's needs. Libraries, student unions and malls require permitted meeting areas and hours, not blanket approval of a building. Campus enrollment must not expose students to staff power imbalances; university geofences can contain homes and medical facilities. Nightlife requires explicit alcohol preference, staffed opening hours, and a non-drinking alternative.

Selection: first filter for both users' accessibility, time, budget, activity, and allowed public-zone constraints. Then prefer low worst-case travel burden, using `max(t_u(v), t_v(v))` and a secondary imbalance penalty. Weights are policy choices to test, not romantic utility. Use approximate origins or on-device route estimates; never transmit one member's origin to the other. Present a stable set of a few zone-level venues rather than changing the midpoint as people move.

Both choose/approve the same venue and start-time version. Either may decline, suggest another approved public option, or cancel without disclosing a personal reason. A venue closure invalidates the plan and requires new bilateral confirmation; no silent rerouting. 'I'm here', 'Running late', and 'I need to leave' are manual messages, not live ETA or routes. Chat and recognizable profile information are available before travel.

An optional trusted-contact share contains only the member-selected venue, time and minimal plan details. Use the OS share sheet or an explicitly addressed contact action; never upload an address book by default. A trusted contact may also be an abuser, so sharing is optional, revocable where possible, and never required to use Go. A future check-in service needs delivery monitoring, escalation rules, and explicit setup. Do not copy v0.1's automatic 'no reply means alert someone' without operational support and false-alarm design.

After a no-show, offer a safe way to leave, support, and any promised fee relief. First check evidence and circumstances; safety exits, transport failures and emergencies are not moral failures. Repeated corroborated abuse of the service can restrict scheduling after review and appeal. No public reliability badge or negative review of a person.

Part 19 - Event/community integration

Events are promising because they concentrate time, place and shared activity. They may also make first contact less intense and permit voluntary familiarity. These are **reasonable inferences**, not proof that event dating produces better relationships than one-to-one introductions.

Start with a few recurring, explicitly opt-in public activities run with known hosts. Examples: a cafe game evening, accessible short walk with staffed gathering point, a class, or a volunteer social. Do not launch a broad event marketplace. An event should have value without a romantic match, and non-dating participants must not be added to a romantic pool.

Data flow: verified organizer creates event and public venue -> operator reviews time, consent language and conduct plan -> member RSVPs -> separate dating-discovery opt-in -> check-in/attendance validation -> temporary event candidate pool -> ordinary bilateral constraints and capacity -> private introduction -> meeting/continuation -> membership expires. RSVP alone neither proves attendance nor authorizes discovery. Event ending revokes Go presence, not a mutually created ordinary conversation.

Use event-specific authorization tokens with scope, expiry, and revocation. Organizers see operational attendance only when necessary; they do not see sexuality, private matches, refusals, feedback, or safety-case details. Do not publish a roster of who enabled Go. No romantic-discovery link to a member's workplace, faith group, or alumni identity without explicit consent.

For Meetup/Eventbrite, MVP can link to reviewed listings or work directly with the organizer. Respect platform terms and partner authorization. Do not scrape attendee identities, infer dating availability from tickets, or assume current APIs permit attendee import. Deduplicate canceled/rescheduled events and require re-consent if the dating scope changes.

Format	Advantage hypothesis	Operational risk	Pilot comparison
Cold one-to-one introduction	Clear intent, low group coordination	More pressure and vulnerability, higher planning cost	Wanted meetings per enrolled member and attention cost
Public event with optional introductions	Concentrated supply and natural conversation	Consent confusion, host quality, outing, group imbalance	Same outcome horizon plus event value without dating
Recurring community activity	Voluntary familiarity over time	Social pressure, repeated unwanted contact, clique formation	Repeat attendance separate from romantic success; enforce pair rejection

Recruit against missing **eligible edges** rather than a simple 50/50 men/women headcount. Capacity, age, orientation, intentions, accessibility, availability and comfort create a compatibility graph. Do not use demographic admission quotas that become exclusionary status rules. Explain waitlists in terms of real operational capacity and provide honest alternatives.

Part 20 - Trust and safety

Safety is a service with accountable people, not a hidden model multiplier. Maintain separate identity assurance, conduct integrity, operational reliability, and safety case management. A concern can trigger immediate precautionary restrictions without declaring an allegation proven; communicate neutrally and provide an appropriate appeal path.

Identity: verify contact ownership and adult eligibility; use proportionate photo/liveness/age checks before high-risk features. Vendor checks need performance evidence, subgroup error review, appeal/manual alternatives, retention contracts, and clear explanations. A verified badge means only the specific check passed. It never means background-checked, harmless, truthful in all respects, or incapable of creating another account.

Conduct: blocks are immediate and bilateral in discovery, with no notice identifying the blocker. Restrict messaging after unmatch but preserve reporting and member access to relevant own records. Monitor enumerating access patterns, repeated creation, abusive invitations, and corroborated integrity problems. Risk flags should add friction or review proportionately, not automatically infer criminality or identity from weak proxies.

Safety operations: staffed review during every live Go pilot window, published support hours, emergency resources appropriate to launch jurisdiction, a trained incident lead, case severity definitions, evidence controls, appeals and escalation. Do not promise 24/7 intervention if it is not staffed. Urgent imminent-harm reports get immediate triage during staffed operations; nonurgent pilot goal is acknowledgement within 24 hours. These are staffing commitments to verify, not guaranteed protection.

Integrity versus reliability: an unverified mismatch can warrant clarification; repeated evidence of deliberate material misrepresentation can warrant review. A late cancellation affects only logistics until evidence suggests abuse. Missing reports or attendance history must not disadvantage new users. Reports are neither public reviews nor features in a romantic model.

Safety tooling and ML: MVP uses transparent rules plus human review. For a bought content/identity classifier, document its actual target, training/evaluation population, error types, appeal rate, and threshold behavior; never assume a vendor's confidence is calibrated for this dating population. A future abuse model would predict adjudicated policy violations in a specified window from permitted operational signals, not 'dangerous person'. It would require sufficiently representative adjudicated positive and negative cases, precision/recall at review capacity, subgroup audits, and manual fallback. Do not auto-dismiss a new report because such a model scores it low.

Contacts: v0.1's 'hashed contacts' are not automatically private: phone/email dictionaries are enumerable. Prefer direct in-app block and manual exclusions initially. A future opt-in contact-exclusion system should use a specialist private-set-intersection design or tightly controlled keyed tokens, strict rate limits, deletion, and no positive match responses. Hashing alone is not anonymization, and a new account can still defeat a known-contact block.

Kill switches: disable a venue, event, account, client version, new-person Go, or all Go independently. Keep reporting and existing safety assistance available during shutdown. A reproducible post-revocation disclosure, exploitable location oracle, or uncontained severe incident stops the affected feature immediately. Restore only after verified remediation and operational review, not because engagement falls.

Part 21 - Threat model

Scope, assets and trust boundaries

Method: STRIDE-style security review plus privacy/physical-harm analysis. Assets include presence, identity-to-location linkage, sexual orientation, private decisions, venue plans, reports, exclusion graph, and account authority. Boundaries: device/OS; authenticated API; live presence process; core database; safety store; analytics/training; staff tools; vendors; public venue and other members. Mobile clients and counterparties are untrusted. Authentication is not evidence of benign intent.

Likelihood ratings are qualitative **pre-control engineering judgments**, not measured probabilities: H = easy/repeatable incentive-compatible attack; M = plausible with some access or effort; L = rarer capability requirement. Severity H = serious privacy/harassment risk; C = credible physical harm or broad sensitive compromise. Residual risk persists even after controls.

Actor / attack	Likelihood / severity	Detection	Prevention	Recovery and residual risk
Malicious user probes cards and filters to identify someone	H / H	Repeated near-identical requests, pair/zone query budgets, unusual retrieval sequences	No arbitrary target/location query, no counts, bounded cards, bilateral gates	Revoke sessions, restrict/review account; already seen identities cannot be recalled
Stalker moves or toggles to triangulate	H / C	Excess toggles, zone hopping, repeated target co-occurrence using minimized counters	No distance bands/direction, sticky session scope, no historical traces, disclosure budget survives toggling	Stop nearby feature for victim on request, investigate, preserve volunteered evidence; physical observation remains possible
Abusive ex uses existing match or new account	M / C	Block evasion signals, targeted repeat patterns and member reports	Recheck blocks every transition, no last-seen/Go status, revocable sessions, manual exclusion options	Immediate protective restrictions and support; new identities and coercion remain hard to detect
Fake account impersonates a real person	H / C	Duplicate/media checks through justified tools, reports, verification inconsistencies	Free proportionate verification, protected recovery, restricted new-account disclosure budget	Suspend pending review, notify affected parties appropriately, appeal; verification can be fooled
Coordinated accounts enumerate or satisfy crowd threshold	M / C	Linked operational behavior, shared devices/network patterns with care for legitimate households	Per-target/pair and zone budgets, verified-account controls, no reliance on k alone	Quarantine cluster, invalidate tokens, reassess disclosure; collusion cannot be eliminated

Actor / attack	Likelihood / severity	Detection	Prevention	Recovery and residual risk
GPS spoofing / rooted device for remote presence	H / H	Impossible jumps with short-lived prior observation, coarse network inconsistency, optional device attestation	No location query without current scoped session, event proof/host check, conservative gating	Revoke Go and request review; attestation/mock-location flags are bypassable and may false-positive
Device theft or coerced unlock	M / C	New-session/device alerts and anomalous actions	App lock/OS protections, reauthentication for sensitive changes, generic pushes, remote session revoke	Account recovery and device revocation; coercive control cannot be solved by an app lock
Compromised backend reads live location	L / C	Credential/access anomalies, egress monitoring, runtime and infrastructure alerts	Isolated presence boundary, least privilege, no persisted raw fixes, short-lived keys, minimized vendor access	Kill Go, rotate credentials, contain, investigate and notify per obligations; live memory remains exposed
Insider searches for a member's movements or reports	M / C	Immutable access audit, unusual staff lookups, dual-control exports	No general staff location UI, just-in-time case access, separated roles/keys	Revoke staff access, independent investigation, scoped notification; legitimate privileged access can still be abused
Historical location database or backup breach	M / C if built	Storage/log scanning, deletion canaries, restore drills	Do not build raw history; exclude presence from WAL/snapshots/logs/backups; minimize venue records	Contain and delete illicit copies where possible; agreed venue/time records remain sensitive historical facts
Queue/cache race leaks after Ghost or block	M / C	Epoch mismatch telemetry, automated race and replay tests	Synchronous policy check at every serve/commit, invalidation, generic push payload, fail closed	Disable offending path, revoke outstanding tokens, incident review; prior delivered card persists
Rendezvous bait or unsafe/closed venue	M / C	Venue checks, user/host reports, plan-change anomalies	Curated public places, stable candidates, bilateral plan versions, no silent rerouting	Cancel, offer exit/support, remove venue; public presence cannot guarantee safety
Insider/outsider joins orientation and event data	M / C	Access graph audits, unauthorized cross-domain joins	Separated audit/safety/identity stores, minimum aggregate sizes, no roster export	Revoke access, breach response, protect affected users; combinations of benign fields can still identify them

Actor / attack	Likelihood / severity	Detection	Prevention	Recovery and residual risk
False or retaliatory reporting	M / H	Pattern review, available evidence, independent reviewers	Confidential reporter access, no automatic permanent ban by report count, proportionate temporary measures	Appeal/correction and retraining exclusions; never require victim confrontation or perfect evidence

Specific release tests

Build adversarial test accounts and synthetic locations in a controlled environment. Verify that moving, filter edits, multiple accounts, cached responses, push timing, venue choice, error text, and repeated Go/Ghost cycles do not create a finer-grained location oracle. Test blocks across devices and stale tokens. Test authorization-store failure and event end. Test that location bodies do not enter traces, crash tools, queues, exports or backups. A nominal UI review is insufficient.

Review with stalking/abuse expertise and users with higher safety exposure, including women and LGBTQ participants, using compensated voluntary research and safe scenarios. No live adversarial tests against unsuspecting members. Residual physical risk and coercion are not permission to abandon the intent; they are reasons to keep Go scoped, consensual and operationally staffed.

Part 22 - Database/data architecture

Service boundaries and entity model

Start with a modular application, managed PostgreSQL/PostGIS, object storage for moderated profile media, a durable job queue/outbox, and an operator console. Event-bound Go can use expiring database session records without GPS. Introduce an isolated nonpersistent presence service for broader Go; it is the justified exception to the single-deployable default. Keep safety access and encryption boundaries real even if infrastructure is shared.

Domain	Entities and essential fields	Invariants
Identity/account	User, IdentityCheck, ConsentRecord: account state, adult eligibility, vendor reference, scopes, versions	Private legal identity separate from display profile; no biometrics in matching
Profile/preferences	Profile, Media, ExplicitConstraint, SoftPreference: value, strength, flexibility, sharing scope, freshness	User edit history preserved; hard gate never rewritten by learned state
Modes/capacity	DiscoveryPolicy, CapacityLedger, Reservation: mode axes, epoch, budget, expiry, owner	Atomic bilateral reserve; exactly-once release; one active Go session per account
Presence	LocationSession, CurrentPresence, GeospatialCell/Zone: opaque token, zone/cell, accuracy class, freshness, expiry	No persistent precise fix or trace; denied when epoch/lease invalid
Pools	CandidatePool, PoolMembership: source, scope, policy, expiry	Overlapping membership; no race/beauty tier; materialize only when needed
Recommendations	CandidatePair, Opportunity, Recommendation, Decision: unordered pair key, mode, state, feature/model version	No duplicate live pair; authorizations rechecked on disclosure; timeout distinct from pass
Coordination	Conversation, MeetNowIntent, Rendezvous, DatePlan, Attendance	Plan requires both confirmations of same version; reports retain provenance
Venues/events	PublicVenue, VenueReview, Event, EventConsent, EventCheckIn	Review expiry, timezone and cancellation; RSVP is not romantic consent
Feedback/learning	MatchingFeedback, BehaviorSignal, PreferenceModel, AttractionModel, ModelVersion	Selection/experience separate; sensitive reports absent; timestamps prevent leakage
Reliability/integrity	ReliabilityEvent, IntegritySignal, ReviewOutcome	Context, evidence, correction and appeal; no global reputation score
Safety	SafetySignal, Report, Block, Enforcement, Appeal, CaseAccessAudit	Separate role/key access; denials available to serving without narrative contents

Domain	Entities and essential fields	Invariants
Measurement	SelectionPropensity, ExperimentAssignment, EventLog, OutcomeObservation, DeletionJob	Versioned policy and denominator; no raw GPS; tracked deletion into exports/models

Use canonical unordered pair IDs for uniqueness, but directional decision rows keyed by member. Version all profiles/features used for an offer. Serialize reservations for both users in a consistent lock order inside a database transaction. The outbox records delivery intent in the same transaction; workers recheck policy before sending. Safety denials must be available synchronously and take precedence over a successful score or old cache entry.

API contracts

`StartGo(zone_or_event, duration, policy_version, consent_version)` returns a short-lived scoped session. `RefreshPresence(session, epoch, current_observation)` returns only acknowledgement, never a nearby user list. `NextOpportunity()` returns at most the allocated permitted card. `Decide(opportunity, version, decision)` is idempotent. `ConfirmPlan(plan, version)` commits only after both current approvals. `StopNearby(expected_version)` advances epoch and acknowledges revocation. `Block(target)` invalidates all discovery paths. `Report(context, optional_detail)` is independent of feedback and remains accessible after closure.

No `/nearby?lat=&lng=&target=` endpoint. No client-readable proximity table, mass profile download, user-search endpoint keyed by phone, or geospatial query in analytics. API authorization is object- and field-specific; predictable IDs cannot authorize access.

Sensitive-field inventory: collection and retention

The following are proposed pilot maxima, with documented exceptions for lawful retention and case-specific evidence. Engineering must verify vendor, cache, backup and log behavior. Unneeded data should be deleted sooner.

ID / data	Why and exact precision	Retention proposal	Could collection be avoided?
D1 Raw GPS fix	Public-zone/proximity check; OS coordinate plus accuracy/time, transient only	Request lifetime, target <=10 seconds; no persistence	Yes for event-bound Go; local-only zone conversion lowers exposure but weakens spoof checking
D2 Current coarse presence	Present eligibility; public-zone ID or H3 r8-class cell, latest timestamp	Overwrite latest; 90-second lease, delete on Stop; session metadata max 24 hours without location	Yes for Explore; event token replaces location
D3 Location history	No ordinary product purpose	Never collected as a history	Yes; explicitly prohibit traces, visited-cell arrays and provider logging

ID / data	Why and exact precision	Retention proposal	Could collection be avoided?
D4 Sexual orientation / who-to-meet	Bilateral consent eligibility; member-declared categories, no inference	While account/setting active; delete active stores within 30 days after request	A separate orientation label is optional; compatible who-to-meet settings still reveal sensitive information
D5 Stated/learned attraction	Selection/experience personalization; low-dimensional posterior with uncertainty	Active use; review inactive state after 90 days; rolling training rows max 180 days	Yes in MVP; do not store visual preference vectors by default
D6 Appearance representation	Optional later experiment, no named race/beauty attributes	None in MVP; if justified, active consent only, <=30-day deletion workflow	Yes; require material benefit before collecting
D7 Culture/religion/language	Optional concrete preference/context, explicitly shared or private	Active setting only; delete within 30 days on request	Usually; language or practice can be asked directly without inferring group identity
D8 Post-date feedback	Private encounter outcome; categorical response, optional chips	Identifiable link 180 days, then delete or truly aggregate after risk review	Optional; outcome measurement can use aggregate research samples
D9 Safety reports/evidence	Protect members, adjudicate, meet legal duties; member-supplied detail only	Proposed 24 months after closure, reviewed annually; shorter if unneeded; case-specific legal holds	Not entirely; collect minimum, never add routine location history 'just in case'
D10 Identity/age/biometrics	Adult eligibility and impersonation friction; retain pass/status/provider reference, restricted birth date if needed	Raw vendor evidence target deletion <=24 hours after completion unless justified contract/legal requirement; token/status while needed	Avoid operator-held ID/face templates; manual alternative where feasible
D11 Profile media/chat	Informed introductions and communication; strip EXIF; ordinary content	Profile while active; chat proposed 90 days after closure unless user-kept or reported; delete active stores <=30 days	Photos needed for chosen design; no matching use of private chat content
D12 Venue/time plans	Agreed meeting coordination; exact public venue and agreed time	90 days after plan closure, then remove identifiable link unless report hold	Needed for plan; not equivalent to continuous whereabouts
D13 Security/abuse signals	Access protection; minimized device token, coarse network/rate counters	Raw access/IP logs 7 days, minimized counters 30 days; justified incidents separately	Minimize; do not reuse as location or identity-model features

ID / data	Why and exact precision	Retention proposal	Could collection be avoided?
D14 Audit demographics	Voluntary fairness research; self-reported categories, optional	Research window max 180 days before renewal/deletion review	Yes; never infer missing demographic answers
D15 Contact exclusions	Block selected known contacts; no address-book ranking	None in MVP beyond direct blocks; later scoped tokens until revoked	Yes; in-app blocking first, specialist protocol if added

Sensitive-field inventory: access, use, deletion and breach impact

All client/server transport uses TLS. Durable sensitive fields use envelope encryption with separated keys and audited service identities; backups are encrypted. Sensitive field access is distinct from routine database access. Live presence has network/process isolation and no durable backup rather than a misleading claim of at-rest encryption for nonexistent files.

Data IDs	Who may access / shown to whom / model use	Deletion behavior and breach risk
D1-D3	Dedicated presence process only; other users see no fix/cell; no recommender training	Stop deletes keys and buffers; no restore. Live compromise can expose whereabouts; logs are the main accidental history risk.
D4, D7	Eligibility service; only explicitly shareable profile facts shown; culture not inferred; model use tightly purpose-limited	Remove settings/caches and future training inclusion; prior disclosures cannot be recalled. Breach can out sexuality or religion.
D5-D6	Restricted training/serving pipeline; owner can reset; never exposed to matches or pricing	Delete personal state/embeddings and samples; rebuild affected model artifacts on a bounded schedule. Breach reveals intimate inferred preferences and can enable discrimination.
D8	Member's own response and restricted learning pipeline; support only when necessary; no automatic mutual-Yes reveal	Delete rows, feature snapshots and export copies. Aggregates must pass reidentification review. Breach risks humiliation and harassment.
D9	Trained safety staff with case-based access and key; serving gets only enforcement/deny status; no romantic-model use	Case-specific deletion/hold review, no indefinite generic exception. Evidence may identify victims and perpetrators; preserve confidentiality and appeal.
D10	Identity vendor/service; reviewer only for a case; others see only precisely described check status	Vendor deletion confirmation, revoke references, remove unnecessary birth date. Breach exposes legal identity/biometrics; templates cannot simply be 'changed'.
D11-D12	Authorized participants, communication/coordination service; safety access when justified	Purge media/CDN caches, plan links, chats per policy; retain reported evidence only in case store. Breach exposes intimate conversations and prior meeting locations.

Data IDs	Who may access / shown to whom / model use	Deletion behavior and breach risk
D13	Security operators; no match visibility or attraction use	TTL deletion, incident hold scoped and audited. IP/device data can still reconstruct identity or movement if retained excessively.
D14	Separate audited research enclave, aggregate output with small-cell suppression	Delete identifiers and joins at expiry; consent withdrawal stops future use. Breach can join protected attributes to intimate outcomes.
D15	Exclusion service only; no positive lookup result shown	Remove revoked tokens; retain direct block protections as justified. Hashes/tokens can expose social ties if keys or dictionaries leak.

Deletion orchestration enumerates primary stores, caches, search, media/CDN, exports, feature snapshots, vendor copies and model state. Proposed backup expiry is 30 days; a restore reapplies deletion tombstones before serving. Remove personal embeddings immediately on the active path; retrain impacted retained global models on a bounded cycle, proposed 30 days, or document and assess an alternative. Do not promise exact machine unlearning without capability. Reports/legal holds remain isolated, purpose-limited, documented and accessible only to authorized staff. 'Append-only' audit logs still need minimization and retention, not indefinite copies of deleted personal data.

Part 23 - MVP recommendation

Phase 0: two to three weeks of a concierge service for roughly 50-100 opted-in adults in one realistically connected local catchment. Use a secure operational tool, minimal profiles and availability, explicit introductions, public venues, basic scheduling, voluntary feedback and staffed support. No precise location collection. Manual matching validates demand and logistics but cannot establish the future algorithm's causal advantage.

MVP: Explore, finite cards, consent/constraints, bilateral capacity reservation, chat/scheduling, public venue roster, reporting/blocking, optional feedback, Focus/Pause/Exit, deletion, and versioned exposure logging. Build event-bound Go as a feature-flagged prototype only after the base service works. Existing-match meet-now is the smallest useful Go test; stranger introductions at a staffed event are a separate consented step.

V1: repeatable local operations, calibrated pooled selection/logistics where data supports them, a better allocation solver if it improves coverage, and bounded neighborhood Go after mobile/privacy validation. **V2:** transferable population models, carefully validated individual residuals, multi-market learning with local calibration, richer event formats. **Long term:** durable-outcome research and selective retrieval embeddings only if the evidence justifies them.

Do not recruit a second market to disguise a weak first one. Do not make sophisticated ML a dependency for launch. Do not make GPS permission a dependency for the entire product.

Pilot service gates

Gate	Proposed evidence before expansion	Failure response
Consent/privacy correctness	No known critical disclosure or revocation defect; adversarial and race tests pass	Stop Go and fix; this is not an A/B variable
Operational readiness	Named incident owner, staffed live windows, reviewed venues, incident/refund playbook, deletion demonstrated	Keep concierge/Explore only
Supply feasibility	At least 80% of enrolled discovery-seeking members can receive one feasible introduction within 14 days, with exclusions and segment uncertainty reported	Adjust recruitment/time/place; no forced boundary relaxation
Meeting follow-through	Target >=60% of mutually confirmed plans reported attended, with unknown and one-sided reports shown separately	Fix coordination/support before adding ranking complexity
Experience	Enough private outcomes to understand discomfort and mutual interest; no severe unresolved recurring pattern	Qualitative review and redesign; no 'zero reports proves safe' conclusion
Go incremental value	Practically meaningful gain over baseline coordination, without worse comfort/privacy outcomes; early target +5 mutually positive meetings per 100 eligible members over 28 days	Keep Go as event experiment or drop it; small pilots may remain inconclusive

These targets are founder decision thresholds, not sourced benchmarks or statistically powered success claims. Freeze them before a pilot and revise transparently from observed base rates. A 12-week pilot cannot demonstrate long-term safety or a durable-relationship lift.

Part 24 - ML evolution by scale

Total users are an inadequate maturity signal. Track local active compatible users, concurrent Go sessions, exposed decisions, matured pair outcomes, feedback response, subgroup support, and feature-serving latency. A million dispersed accounts can support worse Go inventory than a hundred people at one event.

Scale	Rules and population statistics	Personalization / embeddings	Serving architecture and premature work
100 users	Manual-reviewed eligibility, availability, venues, round-robin/greedy capacity. Estimate funnels with wide intervals.	Stated preferences only; no learned personal type or attraction model.	One app/database and secure operator console; event check-in. Feature store, streaming ML and graph clustering are premature.
1,000 users	Rules remain default; pooled base rates and possibly a small regularized selection model if label gates pass.	Heavily shrunk actor selectivity only when supported; experience population summaries. No photo embeddings.	Batch feature tables and authoritative live session/capacity store; no dedicated ML feature store required.
10,000 users	Calibrated joint/selection/logistics models by mode if thousands of matured outcomes exist; benchmark general-graph allocator.	Test small user residuals on well-observed members; most post-date learning still pooled. Collaborative embeddings only if baseline gain is clear.	Separate ephemeral presence if broad Go; batch model scoring plus fresh serving features. A feature platform remains optional.
100,000 users	Retrieval and allocation partitioned by local logistics; population models with regional/mode calibration and drift monitoring.	Test shared embeddings for retrieval if candidate-scoring cost/recall demands it; experienced preference remains strongly pooled.	Online feature store becomes useful if several models need consistent low-latency non-location features. Consent/capacity stay authoritative outside it.
1,000,000 users	Market-level models, robust policy evaluation, heterogeneous effects and mature safety operations.	Selective individual residuals where label-rich; embeddings need continued fairness/utility proof; durable outcomes still sparse/censored.	Regional live-presence partitions, versioned batch/stream features, measured SLOs. Global chemistry engine, universal beauty rank and automatic deep visual models remain unjustified.

Data arithmetic, explicitly hypothetical: 1,000 active members each receiving three introductions per week produce at most 1,500 pair introductions. If 25% become mutual and 50% of those meet, that is about 188 meetings. If 60% of individual attendees answer feedback, expected responses are about 225, not 375. Under an additionally unrealistic independent-response assumption, only about 68 meetings yield both responses. Counts are further reduced by capacity, eligibility and incomplete exposure. Plan sample budgets using observed joint feedback rates, not a hopeful registration count.

At one meeting every two weeks, a member contributes only about 26 meeting experiences a year. That is too little for a rich private model, especially with optional feedback. Population learning, explicit choices, and good logistics will do most of the work for a long time. Use learning curves and confidence intervals, not a fixed events-per-parameter slogan, to decide model complexity.

Deployment gate for any model: mature target definition; valid time-aware data; baseline win; calibration and useful uncertainty; support across key segments; no material coverage/safety regression; rollback and feature provenance; online evidence for the serving policy. Embeddings are not required by a user-count milestone, and a live key-value store for Go is not automatically a real-time ML feature store.

Part 25 - Experiments

Test product mechanisms before model complexity. Do not bundle a cap, new photo layout, event launch and Go into one treatment and then attribute a lift to 'the algorithm'. Eligibility, consent, blocking, revocation correctness and reporting access are fixed requirements, not experiment arms.

Experiment	Hypothesis / design	Primary outcome and guardrails	Decision
Concierge feasibility	People will accept finite introductions and meet. Start with an explicitly opt-in local cohort, log all offers and assistance.	Attended and mutually positive meetings per enrolled member; operator time, coverage, discomfort	Validates demand/logistics, not algorithm superiority
Finite slate/cap	Smaller load improves wanted meetings, not just acceptance. Compare two practical caps in compatible session/cohort blocks after base service stabilizes.	28-day member outcome and concentration; unserved users, perceived pressure, early exit	Choose the smallest cap that sustains good outcomes and agency
Event versus one-to-one opportunity	Shared context improves feasibility. Randomize invitations to eligible opportunity formats where feasible, retain intention-to-treat.	Wanted encounters and meaningful follow-up; event value without romance, cost and outing concerns	Do not compare only self-selected attendees
Go versus ordinary coordination	Immediate availability adds meetings that would otherwise not occur. Randomize eligible staffed event/session clusters with enough repeated units.	28-day outcomes plus same-session meetings; displacement of Explore, safety/comfort, permission refusal	Go ships wider only if practical incremental value is credible
Identity timing	Showing a normal profile before commitment is comfortable and efficient. Begin with simulated usability research, no real blind travel.	Understanding, anxiety, willingness to decline, wasted travel	Keep early identity unless strong evidence supports a safe alternative
Candidate-pool expansion	Adjacent permitted pools recover missed opportunities. Randomize bounded bridge allocations within hard boundaries.	Coverage, meetings and repeat/decline burden; geographic inequality	Expand retrieval, not user exclusions
Learning increment	A pooled model beats rules under the same capacity policy. Shadow first, then controlled serving test.	Calibration plus actual wanted meetings and coverage	No deployment for AUC/accept lift alone

Experiment	Hypothesis / design	Primary outcome and guardrails	Decision
Revenue experiment	A transparent service fee or fixed-term plan supports operations without steering behavior. Test price/value after service is useful.	Paid conversion, refunds, contribution, access disparities and cancellations for safety	No personalized prices or pay-for-rank

Power reality: a conventional two-arm independent comparison of 20% versus 25% outcome rates at 80% power and two-sided 5% significance needs roughly 1,100 independent observations per arm. Dating pairs, repeat users and event clusters are not independent; design effects and missing feedback increase requirements. With few event clusters, a thousand individual responses do not create a thousand randomization units. Treat the early pilot as feasibility and estimation with intervals, not a series of decisive A/B wins.

Marketplace interference is real: assigning one person changes the options and capacity available to others. Research shows experiment bias depends on market conditions and randomization design. [M02] Individual randomization can be valid for isolated interface questions; it is not automatically valid for allocation policies. Prefer market/event-session designs where enough independent units exist, estimate spillovers, and predefine who belongs to the analysis population.

Switchbacks by week are not a cure: a match can persist into future weeks, depletion changes the pool, and relationships do not wash out on schedule. For Go sessions, record repeat participants and carryover; use cluster-robust inference and sensitivity checks. For lasting allocation changes, persistent randomized local cohorts may be preferable if they do not split already sparse supply. If neither design is powered or credible, say that evidence is observational.

Pre-register primary target, unit, horizon, practical threshold, segment guardrails, missing-data handling and stopping criteria. Sequential safety stopping is mandatory; repeated efficacy peeking needs a planned method. Report outcomes for all invited/assigned members, not only those who enabled location or completed feedback. Survey both satisfaction and freedom to say No. Never run an experiment that deliberately exposes someone to a known stalking channel.

Part 26 - Metrics

Use a small outcome dashboard, not a single number that can hide harm. The primary pilot measure is **distinct enrolled discovery-seeking members with at least one mutually positive reported real-world meeting within 28 days, per 100 enrolled members**. Count each member at most once in the period; one qualifying pair can contribute two members. Freeze the enrollment denominator before outcomes, retain nonresponders, and report known-positive as a conservative observed rate rather than the true rate.

This combines early human value with coverage and limits rewards for serial first dates. It remains self-reported, delayed and response-biased. Show mutual second meetings, voluntary durable success exits and attention cost alongside it. A positive meeting is not a certified safe meeting or a relationship.

Metric family	Exact definition / denominator	Interpretation guardrail
Primary observed outcome	Distinct members with both participants privately reporting Yes after a reported meeting / fixed 28-day enrollment cohort	Missing is unknown, not romantic failure; report response bounds and coverage
Follow-through	Both-reported attended plans / mutually confirmed plans; show one-reported, disputed and unknown separately	Never require surveillance or a survey to 'count' a user as worthwhile
Second meetings	Distinct members with mutually reported second meeting / enrollment cohort	Count people once; off-platform meetings and nonresponse create lower bounds
Durable connection	Optional self-reported meaningful relationship exits and 30/90-day continuation among consented follow-ups	No response is not breakup; report denominator and response rate
Coverage	Share offered at least one feasible introduction in 14 days; distribution of time to first offer/meeting	Separate zero eligible supply, retrieval miss, capacity starvation and user pause
Attention cost	Median/p90 active app minutes per reported attended and positive meeting, with cohort totals	Do not record sensitive screen contents; screen-reader use can affect time
Go opportunity	Eligible sessions with an allocated offer / eligible sessions, plus assigned-cohort result	Not a goal to maximize sessions, pings or location permission acceptance
Go funnel	Offers -> mutuals -> plan approvals -> reported meetings -> private positive outcomes	Session and member denominators; cannibalization and event context shown
Accuracy	Reported 'not actually available/near this zone' offers / assessed offers; stale eligibility rejections	User report and lease validity, not continuous tracking ground truth
Privacy agency	Go permission declines, Ghost usage, stop acknowledgement latency and violations	Ghost use is healthy control, not churn to suppress

Metric family	Exact definition / denominator	Interpretation guardrail
Equity	Exposure Gini/top-decile share, waiting distribution, eligibility-conditioned and absolute coverage, slice calibration	Small-sample intervals; investigate structural supply, not only ranker parity
Abuse	Stalking reports, repeat unwanted pair/zone encounters, blocked access attempts, enumeration/spoofing signals, identity issues	Separate raw reports, corroborated cases and operational detections; no target to reduce reporting
Venue safety	Incidents and closure errors per agreed plans/venue-hours with severity	Low reported rate can mean poor reporting or low exposure, not safety
Operations	Time to urgent triage, case backlog, appeals/reversals, vendor errors, deletion completion	Match staffing to actual service promises
Business	Contribution per attended service, per cohort and per acquired member; refunds, organic acquisition and success referrals	Relationship exit is success; revenue cannot justify harmful retention

Report completeness explicitly. For each outcome, show both-observed, one-observed, neither-observed and contradictory responses. Treat safety-related withdrawal as a separate reportable event, not a hidden low-compatibility label. Where missingness could reverse a decision, use bounds/sensitivity analysis rather than imputation presented as truth.

Do not use 'active users' defined by app openings as the only denominator: success, disappointment and algorithmic exclusion all affect activity. Report fixed entry cohorts, user-weeks voluntarily open to discovery, and successful exits together. Safety guardrails include report accessibility and qualitative comfort, not just incident counts. A pilot with no severe reported incidents is still too small to certify safety.

Part 27 - Business implications

The defensible core, if it exists, is local outcome delivery. It consists of trusted recruitment, sufficiently connected local pools, repeatable venue/host operations, a reciprocal allocation policy that serves more people well, and a consented, carefully measured outcome loop. Data becomes an advantage only if it improves decisions and the benefit transfers; sensitive data volume itself is not a moat worth accumulating.

If Hinge copied Go six months later, the toggle, map, venue arrows, timers, and most algorithms would be easy to reproduce. Incumbent density could make its version work better. The startup must win a constrained use case where it reliably creates meetings users prefer, at sustainable acquisition and support costs. Do not claim that incumbents cannot change incentives or learn from dates.

V0.1 is right to examine incentive alignment but too categorical about pricing. Monthly subscription revenue has a simple direct dependence on duration; referrals, reputation, lower dissatisfaction and acquisition efficiency can counteract it. Pay-per-date rewards first-date volume and can also misalign with lasting relationships. A fixed-term package can encourage under-service or repeated renewals. No pricing formula automatically makes the company benevolent.

Test a transparent, non-renewing local service package or a small coordination fee with clear credits/refunds, alongside ticketed hosted activities if operationally justified. No boosts, extra visibility, hidden paywalls on safety, outcome-dependent pressure, or fees to report/cancel for safety. Never guarantee a number of dates in a way that pressures someone to accept; a service guarantee can cover scheduling effort or refund terms instead.

Illustrative unit economics, not market facts: if each participant pays $12 for a delivered meeting service, revenue per pair is $24. If payment/support/verification allocation/venue subsidy and hosting cost total $18 per pair, contribution is $6 per pair or $3 per participant. A participant attending three times contributes only $9 before fixed costs and acquisition. At $25 CAC, that scenario is unviable even with a popular product. Replace each input with actual pilot costs; include incident handling, refunds and founder labor rather than pricing them at zero.

Venue partnerships should provide access, service or perks without buying recommendation rank. Disclose commercial relationships. An expensive venue must not win over an accessible one because it pays. Keep free/non-purchase meeting options where possible and measure whether fees disproportionately exclude members with lower incomes.

Measure whether success brings voluntary referrals, but never require users to share relationship status publicly or invite friends to unlock opportunities. The strongest retention outcome may be a user leaving. Operating plans should distinguish happy exits, breaks, dissatisfaction and unknown exits rather than treating all as churn to reverse.

Go's capital implication: citywide concurrency, mobile reliability and safety operations are expensive before they generate revenue. Event-bound Go purchases concentrated supply with operations effort; it is cheaper to test but not free to scale. Prove contribution and repeatable local activation before building broad location infrastructure or expanding geographically.

Part 28 - V0.1 -> V0.2 changelog

References to OLD ASSUMPTION below are grounded in the supplied 31-page v0.1. FOUNDER REVIEW distinguishes the revised brief from new corrections made in this review.

OLD ASSUMPTION	FOUNDER REVIEW	YOUR ANALYSIS	V0.2 DECISION	WHY
Dealbreakers are binary and always override learning (pp. 7-9).	Separate true constraints from flexible declarations.	A user-labeled boundary cannot become soft secretly.	Explicit hard / strong-flexible / soft taxonomy; migration defaults hard.	Preserve agency while learning permitted preferences.
Shared future two-hour window in ten days (p. 14).	Add availability now.	One mandatory window excludes valid logistics.	Explore flexible future planning; separate Go lease and immediate availability.	Different timescales need different feasibility tests.
Avoid live location; neighborhood origins only (pp. 18-19, 28).	Investigate private real-time proximity.	Ephemeral processing is possible; gradients still leak.	Event Go first; isolated expiring presence later; no person radar.	Retain intent without building movement history.
Weekly curated introductions dominate (pp. 19-20).	Explore + Go + events.	Multiple modes need coherent opportunity semantics.	Connection Opportunity with type-specific policy.	Shared consent/coordination, distinct inventory.
City/cohort density (pp. 1, 6).	Dynamic local/social clusters.	Permanent segmentation can exclude and run dry.	Overlapping logistics/event pools with permitted bridges.	Improve feasibility without social tiers.
Two or three explanatory sentences (p. 20).	Minimal frontend; human discovery.	Too little identity harms informed choice.	Short factual context plus adequate photos/intention before commitment.	Simple experience without blind travel.
Learn attraction from decisions (p. 8).	Develop uncertainty-aware attraction.	Accept is not attraction or chemistry.	Separate selection and experienced attraction; no visual model in MVP.	Valid targets and sparse-data restraint.
Private reciprocal estimates, no tiers (p. 12).	Solve 'league' with pair probabilities.	Correct principle; hidden popularity can still dominate.	Preserve pair-specific use; no global ladder, pricing or tiers.	Dignity and controllable exposure.
Product of stage probabilities; blindness supports independence (pp. 12-14).	Critique formula.	Blind decisions can share unobserved causes.	Correct conditional chain; joint acceptance model when supported.	Avoid false probability precision.
Rule score scaled 0-1 at MVP (p. 28).	Require real algorithm.	Scaled agreement is not calibrated probability.	Label heuristic explicitly and compare to pooled baselines.	Honest model interpretation.

OLD ASSUMPTION	FOUNDER REVIEW	YOUR ANALYSIS	V0.2 DECISION	WHY
Not sure = 0.5, lower reliability (p. 10).	Retain private three-way answer.	Unsure is a distinct state, not a half-success.	Categorical label and separate missingness.	Avoid invented statistical meaning.
Yes down-weighted if no second date in 21 days (p. 10).	Stronger real outcomes.	Off-platform activity and logistics invalidate the inference.	Preserve Yes; observe later outcomes separately with censoring.	Avoid silently relabeling experience.
One required feedback tap (p. 9).	Tiny interaction feedback, independent safety.	Required responses create pressure and selection artifacts.	Optional/skip; no access or refund consequence.	Consent and better measurement.
Mutual Yes feedback revealed (p. 10).	Minimum private information.	A feedback answer is not communication consent.	Separate 'Invite again' action.	Keep the privacy promise meaningful.
Bayesian per-member online update (pp. 10-11).	Confidence, decay and sparse-data caution.	Early personalized parameters are weakly identified.	Pooled regularized batch models, then shrunk residuals.	Easier audit and honest uncertainty.
Thompson sampling plus one-in-five random slot enables unbiased replay (pp. 14-15).	Preserve propens ities/exploration.	Allocator propensities and missing outcomes complicate causal claims.	One explicit randomized policy; conditional probability logging; supported estimands only.	No blanket debiasing promise.
Min-cost flow suggested alongside b-matching (p. 15).	Capacity/exposur e fairness.	General eligibility graphs need general-graph treatment.	Greedy MVP, benchmark general b-matching; flow only when bipartite.	Inclusive and mathematically valid allocation.
At least one offer per fortnight if score clears threshold (p. 15).	Avoid attention concentration.	Floors can conflict and become infeasible.	Soft coverage deficits and transparent unserved reasons.	Never force consent or overbook.
Hashed contact upload (p. 16).	Strong privacy and abuse design.	Hashing low-entropy identifiers is enumerable.	Direct blocks first; specialist private exclusion protocol later.	Protect social graph and account presence.
Mandatory vendor liveness; 'shows up' badge; two no-shows suspend (pp. 16-17, 28).	Separate legitimate behavior from moral inference.	Verification has errors; badges stigmatize new users; no-shows need context.	Proportionate free checks, appeals, private contextual reliability.	No false safety certificate or simplistic punishment.
Minimize days, not minutes (p. 19).	Go permits immediate meetings.	Same-hour meeting can be wanted with adequate choice and support.	Test immediate coordination in public staffed contexts.	Preserve option without pressuring speed.

OLD ASSUMPTION	FOUNDER REVIEW	YOUR ANALYSIS	V0.2 DECISION	WHY
Trusted contact automatically alerted after no reply (p. 18).	Borrow safety coordination.	Requires explicit setup, reliable delivery and false-alarm handling.	Manual share first; later operationally supported check-in.	Do not promise an emergency system we lack.
Second dates per active users is North Star (pp. 21-22).	Reevaluate meaningful outcomes.	Active denominator and repeat dates can be gamed.	Distinct-member fixed-cohort outcomes plus second meetings and exits.	Coverage and attrition remain visible.
Member A/B invalid; week/city randomization (p. 23).	Rigorous experimentation.	Interference depends on treatment; switchbacks have carryover.	Choose design by estimand/supply; use cluster and carryover analysis.	Avoid false causal confidence.
Pay-per-date preferred; monthly subscription rejected (pp. 23-25).	Align business with relationships.	Every model has incentive risks.	Test transparent service pricing; prohibit paid ranking/safety barriers.	Evaluate full economics and behavior.
One monolith and one database (p. 27).	Add location/privacy pipeline.	Mostly sound, but live presence and safety need stronger access boundaries.	Modular core plus isolated ephemeral presence when needed.	Complexity only where the threat model justifies it.
Market announcements and reported growth motivate differentiation (pp. 1, 30-31).	Research current competitors.	Announcements, availability and independent outcomes differ.	Current official feature snapshot; omit unrevalidated market numbers.	Do not anchor strategy on stale or unsupported claims.

Part 29 - What NOT to build

Do not build during the first 12 weeks:

- Citywide stranger radar, near/nearer meters, person directions, exact location sharing with dates, crossing histories, AR people markers, or counts of compatible people in a small area.
- Background movement history, home/work inference, contact-graph ranking, camera-roll ingestion, or a precise-location analytics warehouse.
- Race/caste/beauty clusters, global desirability ranks, leagues, visual-attraction scoring, or inferred sexuality/religion/personality/morality.
- A long onboarding battery, a chemistry percentage, public reviews, attendance badges, mandatory post-date surveys, or mandatory sharing with a trusted contact.
- Infinite feeds, like inboxes, streaks, loot-box mystery, refresh rewards, fake nearby signals, paid exposure, or a requirement to reply to unlock safety/access.
- A custom feature platform, two-tower retrieval, deep multimodal models, online Bayesian personalization, graph community detection, or many predictive models before labels exist.
- Open event marketplaces, venue booking integrations, live partner ETA, AI-authored romantic conversations, or an unsupported emergency-response service.
- A second city to mask local failures, broad paid acquisition before contribution is known, or outcome guarantees that pressure acceptance.

Do not remove useful friction merely because a model can automate it. Identity inspection, a question before travel, bilateral plan confirmation and easy cancellation are product value. Do not add friction to report, block, pause, leave, revoke location, or delete data.

Refuse the claim that any missing piece is solved by 'AI'. A model that cannot name its outcome, observations, bias, uncertainty and baseline is not ready for the architecture.

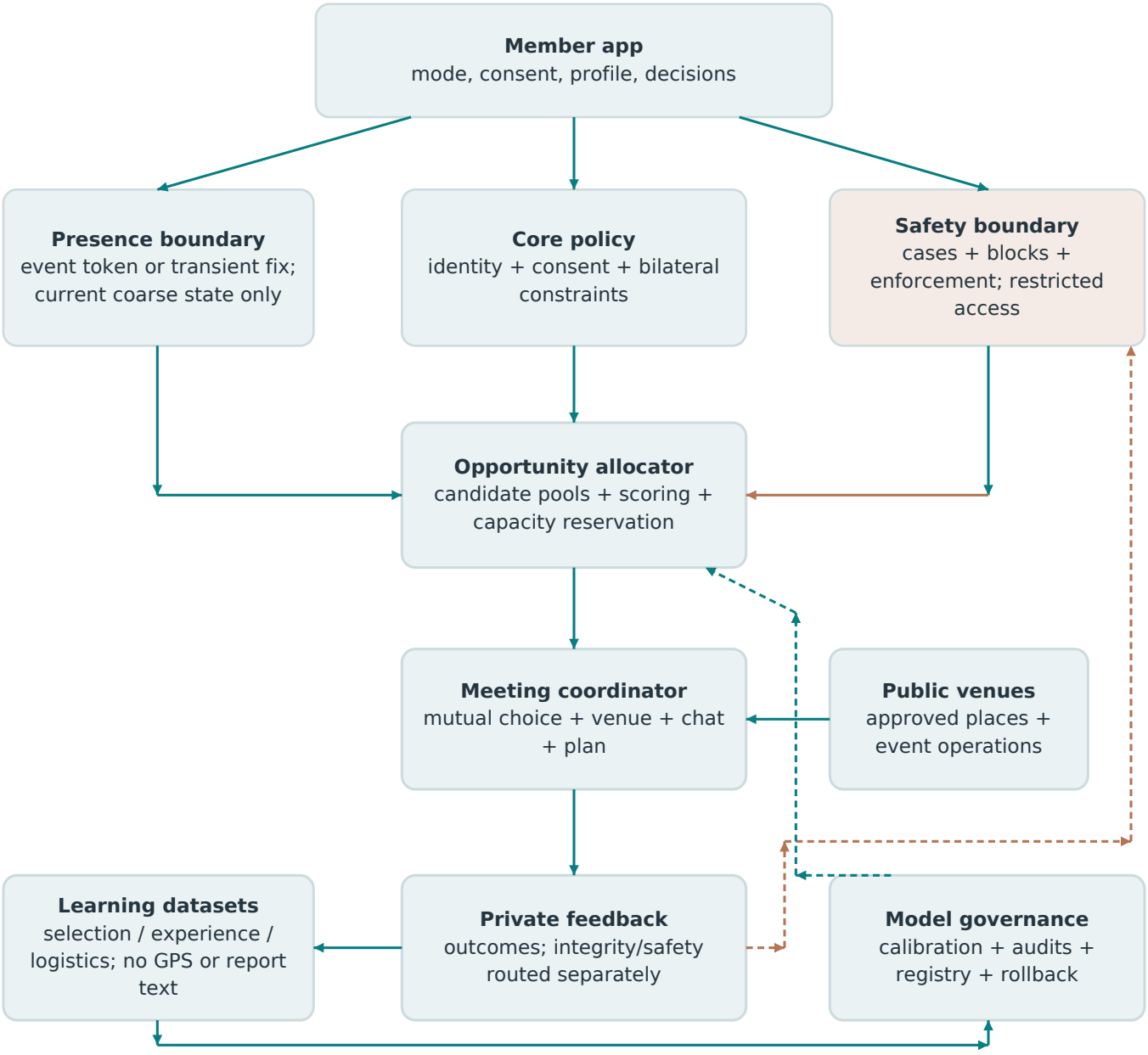
Part 30 - Pre-mortem

Assume the company fails catastrophically after three years. These are concrete mechanisms to watch before growth hides them. Thresholds should be set from pilot baselines; critical privacy/safety defects trigger action regardless of aggregate rate.

Failure mode	Early-warning metric or observation	Mitigation / decision
1. Local supply is too thin	High zero-feasible-edge share and long offer waits despite registrations	Concentrate catchment/time; event sessions; stop expanding cities
2. Women or other high-risk users disable Go	Cohort-specific Go refusal, low voluntary return, safety interviews	Redesign with affected users; keep Explore viable; do not pressure opt-in
3. Location fear prevents adoption	Permission abandonment and misunderstood privacy controls	Event tokens/no-GPS route, precise explanations, shorter leases
4. Radar becomes creepy	Reports of being followed, repeated checking, discomfort before meeting	Remove all gradients/counts; venue-first coordination; stop affected feature
5. Novelty fades	Outcomes fall after first sessions while check-ins remain	Evaluate repeat-cohort outcomes; keep only coordination value
6. Matching adds no value	No lift over simple rules on calibrated held-out and online outcomes	Compete on operations; stop personalization spend
7. Over-clustering hides good options	Strong random bridge performance; high retrieval false negatives	Overlapping pools, permitted expansion, independent retrieval audit
8. Racial or class proxies dominate	Exposure/coverage disparities unexplained by consented feasibility; sensitive probes	Remove features/model, audit eligibility and geography, restore exploration
9. Capacity starves less-selected users	Rising waiting-time tail and concentration despite caps	Waiting-time allocation, new-user priors, improve local supply; no forced likes
10. High-demand users feel rationed	Repeated premature closes, frustration about slot limits	Clear capacity semantics, lower pressure, tune cap without selling bypass
11. Battery/network burden is excessive	Battery delta, crash rate, stale leases on older devices	Foreground/event mode, sparse fixes, graceful expiry; defer ambient Go
12. GPS errors create false opportunity	Wrong-zone complaints and accuracy failures	Accuracy gating, host check-in, public polygons, no false precision
13. Spoofing and Sybils defeat safeguards	Account churn, zone hopping, coordinated query patterns	Scoped budgets, proportionate verification, staffed events, adversarial tests
14. Ghost has race-condition leaks	Any successful old-epoch disclosure after acknowledgement	Immediate kill switch; repair authorization path and invalidate sessions

Failure mode	Early-warning metric or observation	Mitigation / decision
15. Location histories leak through vendors	GPS in logs/crash traces, excessive vendor retention	Minimized payloads, SDK review, deletion canaries, contractual and technical controls
16. No-shows destroy trust	Disputed attendance, repeat last-minute plan failures	Better confirmations/logistics and review; no automatic moral score
17. Unsafe venue or rendezvous incident	Venue complaints, expired checks, closed-place suggestions	Remove venue, assist affected people, staff review and plan reconfirmation
18. Cohort imbalance grows	Unserved eligibility components even with balanced headcount	Recruit for actual graph gaps and time availability; transparent waitlist
19. Events become low-quality inventory	Attendance without wanted interactions; host complaints	Fewer curated formats, host standards, no metric credit for ticket clicks
20. Events expose dating/sexual identity	Roster leakage, unwanted organizer knowledge, outing reports	Separate consent, no romantic roster, restricted organizer access
21. Feedback becomes coercive or gamed	Suspicious uniform Yes patterns; complaints about surveys	Optional feedback, no rewards/gates, separate invitation action
22. Silent/off-app outcomes poison models	Feedback response shifts; predictions degrade by context	Unknown/censored states, consented research panel, no absence-as-negative
23. Safety team is overwhelmed	Urgent queue age, inconsistent decisions, appeal reversals	Limit live windows/growth, train and staff before expansion
24. Pricing rewards bad volume	More first dates but worse satisfaction/exits; high safety-cancel penalties	Remove pressure incentives, transparent credits and service pricing
25. Acquisition economics fail	Negative contribution after support/refunds and founder labor	Community recruitment, pricing/format iteration; stop paid scale
26. Regulatory/vendor obligations exceed capability	Unmapped data flows, deletion failures, consent complaints	Reduce sensitive collection, legal/vendor assessment, delay markets/features
27. Incumbent copies Go and wins on density	Users prefer competitor despite similar UX	Demonstrate local service quality and outcomes or narrow to a viable niche
28. Success metric hides excluded users	Primary rate rises while enrollment shrinks or silent exits grow	Fixed denominators, unserved coverage and exit reasons; audit enrollment policy
29. Community accountability becomes coercion	Pressure to accept, gossip, repeat approaches after No	Private decisions, strict pair exclusion, optional contexts, remove abusive hosts
30. Investor narrative forces premature ML	Many scores, few matured labels, no baseline advantage	Require model contracts and gates; prioritize service evidence over demos

Part 31 - Recommended architecture diagram



Read the boundaries: the core receives an ephemeral eligibility result, not a location trace. Safety sends enforcement decisions to serving but not report narratives to learning. Users' private outcomes enter stage-specific training only under the applicable data policy. The learned policy returns to allocation through validation and audit; it cannot change consent or safety gates.

The MVP's main application contains identity/consent, profiles, eligibility, candidate queries, capacity allocation, opportunity lifecycle, messaging, plans, venues, feedback routing, notifications and the operator console. Use a transaction/outbox boundary for reliable state changes. Event-bound Go can initially run

through these modules with scoped event tokens. A separate live-presence service is added only for neighborhood Go.

The diagram's learning branch contains three distinct datasets: selection exposure/decisions, experience outcomes, and operational reliability. Integrity/safety cases are outside it. An approved model registry stores training snapshot, feature allowlist, label definition, validation/calibration reports, slice audits, intended mode/market, owner, rollback version and deletion lineage.

Graceful degradation: missing model -> rules; missing feature -> explicit default/unknown; no capacity -> no new proposal; stale presence -> Ghost/nearby unavailable; failed authorization check -> deny; unavailable maps -> previously reviewed static venue choice only if still valid; failed venue validation -> no plan; unavailable safety coverage -> no live Go session. Reliability means a safe failure mode, not always displaying somebody.

Part 32 - If I were your technical cofounder, this is what I would build

I would build a locally concentrated introduction and meeting service that works with a simple allocator. I would run it closely enough to learn why people decline, fail to plan, meet, or want to meet again. I would add a narrow Go experiment only after that service works. I would not spend the first quarter predicting visual types or designing an ambient city radar.

The next 12 weeks

Planning assumption: two experienced product engineers, a founder owning product/research, a founder or operator owning recruitment/venues/safety, and part-time design plus privacy/security/legal review. This is a workload assumption, not knowledge of your actual team or budget. If those people are unavailable, reduce scope to concierge Explore and event check-in; do not remove safety work to preserve Go.

Weeks	Concrete work and owner	Reviewable output / exit condition
1-2	Product/ops: recruit 50-100 opted-in adults in one catchment, interview different risk/eligibility groups, review 10-20 public venues, map data flows and staffing. Engineering: consent/constraint prototype and threat tests.	Cohort eligibility graph, privacy/incident plan, working prototype, secure concierge workflow. Proceed only if feasible edges and operational support exist.
3-4	Ops: deliver manual finite introductions, assist scheduling, collect optional outcomes. Engineering: account/profile, explicit boundaries, block/report, capacity ledger, plans and exposure logs.	End-to-end first meeting path with provenance; test state races/deletion; decide which logistics failure matters most.
5-6	Engineering: ship private Explore pilot with finite cards, mutual chat, public venue confirmation, feedback routing, Focus/Pause/Exit, operator tools and outbox. Product: observe onboarding and cancellation.	Reliable introduction-to-meeting flow, no critical permission leak, truthful empty state, staff able to handle cases.
7-8	Engineering/ops: scoped event sessions, expiring check-in tokens, Go/Ghost controls, existing-match meet-now, consented plan sharing. Security: revocation/collusion tests.	Staffed event Go prototype, no person-location output, shutdown and offline behavior demonstrated. No neighborhood GPS requirement.
9-10	Product/data: compare event coordination formats where feasible; inspect full funnel, missingness, coverage, costs and comfort. Engineering: fix identified friction; shadow pooled model only if label gates met.	Descriptive outcome/interval report, actual support cost, candidate coverage audit; no unpowered victory claim.

Weeks	Concrete work and owner	Reviewable output / exit condition
11-12	Team: evaluate continuation gates, test limited willingness to pay for a delivered service, review incidents/appeals and referrals. Decide V1 scope.	Written continue/narrow/stop decision, updated retention/deletion checks, measured roadmap. Broad Go remains conditional.

A thin responsive app or one shared mobile codebase is enough for this pilot. Event-bound Go avoids making background-native location the critical path. Build only the native location capability needed for a later, approved neighborhood experiment, with actual device testing. Do not choose one OS without checking the recruited cohort's access; exclusions caused by platform choice are part of the pilot's limitations.

Week-12 decision: continue if real local demand, feasible coverage, wanted meetings, manageable safety operations and plausible contribution are present. Narrow to events if they carry the value. Keep Go experimental if evidence is inconclusive. Stop or redesign if the product creates anxiety, cannot serve a meaningful fraction of the recruited pool, or requires unaffordable acquisition/coordination. More features are not the default response to weak demand.

Direct answers to the founder's final questions

Does Go fundamentally improve the company, or is it merely a compelling feature? Today it is a compelling, unvalidated coordination feature. It could materially improve the company if it repeatedly creates additional mutually wanted meetings with acceptable operational cost and privacy risk. Its broad ambient version could also make the company worse by increasing fear, empty sessions and safety burden. Event-bound Go is the appropriate test, not a commitment to the full concept.

What is defensible if Hinge copies it? Not the toggle, haptic, map, venue routing, embeddings, or 'designed to leave' story. The plausible defensible core is a trusted local network and operating system that consistently turns reciprocal intent into good real-world experiences, together with a demonstrably useful, consented outcome-learning policy. If that advantage does not show up in outcomes and economics, there is no proven moat yet.

What would I refuse to build with your money? A people-tracking interface, a hidden racial/beauty sorting system, a compatibility oracle, an engagement casino, or infrastructure justified by hypothetical future scale. I would buy evidence through real introductions, good operations and careful measurement. The first technical achievement should be that two people can freely say Yes, meet with less friction, say No without cost or exposure, and leave the service when it has done its job.

Sources and evidence notes

Research and documentation were checked on 11 October 2026. Historical studies retain their original date and population; this report does not relabel them as current causal evidence. Official product pages are company descriptions, not independent validation. Source IDs in the report link claims to this register. Proposed architecture, formulas, settings and judgments are the author's synthesis unless stated otherwise. No external dataset, proprietary algorithm, production app binary, or interview study was independently audited for this report.

Supplied sources: Founder brief, Pasted text.txt, sections 1-90 plus closing clarifications. Product Architecture v0.1 - The Learning Loop, 11 October 2026, 31 pages. V0.1 is used as the design baseline, not as independent proof of external claims. Unrevalidated figures about competitor growth, revenue, funding, future launches and company outcome lifts are not carried forward as facts.

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[L01] Illinois General Assembly. Biometric Information Privacy Act, 740 ILCS 14, especially section 15. Official statutory text; applicability and current amendments require launch counsel review. www.ilga.gov/legislation/ilcs/fulltext?DocName=074000140K15

[L02] California Department of Justice. CCPA FAQ, updated 28 August 2026. oag.ca.gov/privacy/ccpa

[L03] FTC. Final orders concerning X-Mode/Outlogic (April 2024) and InMarket (May 2024). These are case-specific enforcement actions. www.ftc.gov/news-events/news/press-releases/2024/04/ftc-finalizes-order-x... and www.ftc.gov/news-events/news/press-releases/2024/05/ftc-finalizes-order-i...

[L04] European Union. Regulation (EU) 2016/679 (GDPR), including Articles 5, 6, 9, 25 and 35. eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32016R0679

Open questions to resolve with the pilot

The launch catchment, team capacity, operating budget, initial eligibility mix, support hours, event partners and willingness to pay are not supplied. This report makes explicit planning assumptions rather than inventing them. The strongest unresolved scientific/product questions are whether finite attention increases good meetings; whether Go creates incremental value; whether recurring context improves outcomes without pressure; and whether experience-informed learning beats simple feasibility and reciprocal-choice baselines. Those questions determine the next investment, not the attractiveness of the demo.